



POLITECNICO
MILANO 1863

SCUOLA DI INGEGNERIA INDUSTRIALE
E DELL'INFORMAZIONE

Rolling-Horizon A^* Planning and Nonlinear MPC for Map-Free Navigation of a Legged Robot

PROJECT REPORT

062047 – NUMERICAL OPTIMIZATION FOR CONTROL

Professor:

Lorenzo Mario Fagiano

Group Members:

Lorenzo Ortolani - 10...

Francesco Pedrini -
10787879

Academic year:

2025-2026

Abstract: Autonomous navigation in an unknown, cluttered environment can be posed as a pair of nested optimization programs solved at every control cycle: a discrete shortest-path problem on a LiDAR-derived occupancy grid, and a continuous nonlinear program (NLP) that turns the resulting geometric reference into dynamically feasible velocity commands. Here, we analyze the map-free navigation of a legged robot, a Unitree G1 humanoid. The high level plant consists in a non-linear forward kinematic dynamics, valid for each legged robot abstracted as a body-frame holonomic agent, with first-order actuator lag for modeling the low level dynamics latency, yielding a six-state, three-input discrete model; the nonlinear MPC tracker formulated in CasADi and solved by IPOPT at 8 Hz. The eleven cost weights of that NLP are firstly tuned by heuristics, but can be further improved by an offline calibration using a per-axis Bayesian scheme: a one-dimensional Gaussian process per weight, six closed-loop mission rollouts each, swept in order of influence with every optimum carried forward, for 67 rollouts in total. Every claim in the performance analysis is produced by driving the deployed CasADi/IPOPT tracker in closed loop inside an offline harness, over different synthetic worlds and many parametric studies. Every design decision in the project derives from theoretical considerations and quantified outcomes: the integration method to discretize the model and the derivative computation approach which results more efficient, the Hessian computation method to be employed, the decision between interior-point and active-set solvers, sparse against condensed parametrisations, a smooth penalty against an exact ℓ^1 relaxation of the clearance constraint, a time-parametrised against a path-parametrised reference, and a terminal equilibrium constraint against none. The first-order and second-order optimality conditions are verified on recorded missions rather than assumed.

Key-words: Nonlinear Model Predictive Control, Interior-Point Optimization, Rolling-Horizon Planning, Bayesian Optimization, Legged Robot Navigation

Contents

1 Introduction

2

2	Problem analysis	3
2.1	System overview	3
2.2	The controlled platform	3
2.3	Low-level locomotion layer	5
2.3.1	Learned whole-body control: the AMO policy	5
2.3.2	The cascade and its abstraction seam	6
2.4	Environment representation	6
2.5	Plant model and state-space formulation	7
2.6	Navigation problem statement	8
3	Formulation of the optimization program	8
3.1	Reference generation: A^* as a discrete program	8
3.2	Model discretization	11
3.3	The nonlinear MPC program	12
3.3.1	Decision variables, cost, and the parametrisation of the program	12
3.3.2	Obstacle avoidance as a smooth penalty	13
3.3.3	The obstacle as a true constraint: exact ℓ^1 penalty	15
3.3.4	Constraints	16
3.3.5	Terminal ingredients and recursive feasibility	16
3.3.6	Reference trajectory: time-parametrised against path-parametrised	17
3.3.7	Robust constraints: tightening from the measured prediction error	18
3.3.8	On the absence of integral action	19
4	Optimization procedure	19
4.1	Problem dimensions and structure	19
4.2	Derivative computation	20
4.3	Solver methodology	20
4.4	Interior point against active set	21
4.5	Optimality conditions on the deployed problem	21
4.6	Implementation and configuration	22
5	Performance analysis	23
5.1	Experimental setup	23
5.2	The cost landscape, and why this is not a potential field	23
5.3	Prediction error: the model against the plant	24
5.4	Regularity of the solution and the left–right bifurcation	25
5.5	Prediction horizon and sampling time	27
5.6	Control horizon, and why not move blocking	29
5.7	Multi-objective scalarisation	30
5.8	Offline weight tuning by Bayesian optimization	32
5.9	Non-convex obstacles, and the target rule that fails on them	33
5.9.1	The limit cycle	33
5.9.2	The two repairs, and what each of them fixes	33
5.10	Summary of design guidelines	35
6	Discussion and limitations	35
7	Conclusions	37
	Reproducibility	37

1. Introduction

A mobile robot that must reach a goal in an environment it has never seen has no global map to plan on: at every instant it knows only what its sensors reported in the last few metres. The classical answer is to decouple the problem into a geometric planner and a low-level tracker, but if the two layers ignore each other the planner produces paths the actuators cannot execute, and the tracker produces commands the geometry does not allow. Model Predictive Control (MPC) resolves the conflict by construction: it optimizes a finite sequence of future inputs subject to the actual dynamics and the actual constraints, and re-optimizes at every cycle [7]. The price is computational for the motion planner point of view: a nonlinear program must be solved to sufficient

accuracy inside the sampling period, every period, on the robot. However, this rolling horizon path planning, allows the robot to avoid simultaneously localize and map the environment. The presence of only a localization module through an EKF and the lack of complete SLAM, enables less computationally expensive navigation stack, which is a core contribution of this project.

The goal is to analyze the navigation stack of a Unitree G1 humanoid. The locomotion layer of that machine, the AMO reinforcement-learning policy [4], exposes a body-frame velocity interface, and it is that interface — not the twenty-nine joints underneath it — that the controller of this report addresses: the high level plant is a simple non-linear forward kinematic model, valid for a legged robot abstracted as a body-frame holonomic agent with first-order actuator lag standing in for the latency of the low level dynamics. Section 2.3 describes that locomotion layer and the abstraction seam at which the optimization program stops seeing it, because that abstraction is the hypothesis on which the whole prediction rests. Two programs are solved at every planning cycle and are the object of the analysis:

- a *discrete* shortest-path program — an A^* search over a Gaussian occupancy grid rebuilt from the current LiDAR scan, whose solution is a geometric reference and whose rolling-horizon target is the intersection of the line from the robot to the global goal, with the grid boundary;
- a *continuous* nonlinear program — a six-state, three-input MPC with actuator-lag dynamics, a smooth obstacle barrier and box-constrained inputs, formulated once symbolically in CasADi [2] and solved by the interior-point method IPOPT [11] at 8 Hz.

Moreover, other two logics are implemented: the first one is needed to deal with the complex scenario which may be encountered during navigation, specifically non-convex obstacles; the other is the offline program which chooses the eleven cost weights of the MPC exploiting Bayesian optimization. Weight tuning is the practical bottleneck of any deployed MPC, it is not differentiable through the closed loop, and each evaluation costs a full navigation rollout. It is therefore attacked with Gaussian-process surrogates [6, 10], but *one per coordinate* rather than one over all eleven, for reasons of sample density that Section 5.8 makes precise.

2. Problem analysis

2.1. System overview

The stack is a chain of ROS 2 nodes that closes the loop between a 3-D LiDAR and the body-frame velocity interface of the locomotion controller:

1. an **odometry bridge** that converts the extended-Kalman-filter (EKF) pose estimate into a single pose topic and differentiates it, with an exponential filter, into a body-frame velocity estimate (the robot does not publish a usable body-frame twist);
2. a **mapping module** that rebuilds a local Gaussian occupancy grid from the newest scan at 1 Hz and maintains a sparse world-frame cell map of confirmed obstacle cells (with a 20 s decay), on which the memory-augmented target rule of Section 5.9.2 plans; its purpose is precisely to repair the failure mode analysed in Section 5.9.1;
3. the **A^* local planner**, which solves the discrete program of Section 3.1 on that grid;
4. the **nonlinear MPC tracker**, which solves the NLP of Section 3.3 at 8 Hz and publishes a setpoint;

Two design decisions shape everything that follows. First, the locomotion layer is treated as an ideal velocity-tracking device with a time constant: the report never models legs, contacts or gait, and the robot is abstracted as a planar holonomic agent, Sections 2.2 and 2.3 describe what that layer actually does on the G1, and what the abstraction discards. Second, the map is *local and volatile*. The occupancy grid covers a fixed 12×12 m window centred on the robot and is rebuilt from scratch at every planning cycle, so the planning problem is genuinely map-free, and the geometric reference the MPC tracks is only valid for one grid width ahead. Section 5.9.1 measures what this choice costs, and Section 5.9.2 what it takes to repair it.

2.2. The controlled platform

The stack drives a Unitree G1 humanoid (Fig. 1). The G1 has twenty-nine actuated joints, of which the locomotion layer of Section 2.3.1 drives fifteen, the twelve leg joints and the three waist joints — while the arms are held at a fixed posture. That joint set is never addressed by the controller of this report. The platform is commanded through a single channel, a body-frame twist $u = (v_x^{\text{cmd}}, v_y^{\text{cmd}}, \omega^{\text{cmd}})$, and carries a 3-D LiDAR whose returns are the only exteroceptive input of the stack. Table 1 collects what the controller sees of it; everything else about the robot is, by construction, invisible to it.

The admissible input set is the row to keep in mind for the rest of the report, and it is a property of the *policy* rather than of the mechanism. AMO is trained inside an envelope of 0.5 m/s and 0.4 rad/s; commanding the robot outside it is not aggressive tuning but a modelling error, because the optimizer would then plan

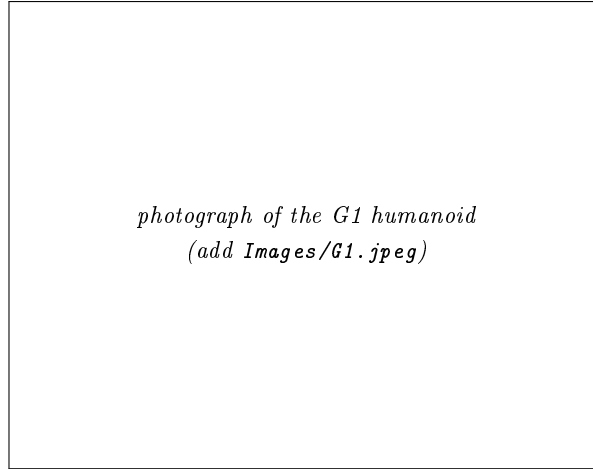


Figure 1: The Unitree G1 humanoid, the platform on which the navigation stack analysed here is deployed. The controller never addresses a joint: it publishes a body-frame twist, and the locomotion layer of Section 2.3.1 turns that twist into the fifteen joint commands that produce the walk.

Table 1: The platform as the controller sees it. The rows above the first rule are properties of the machine, those between the rules are properties of the locomotion layer that receives the MPC command, and those below are the only ones that enter the optimization program: the admissible input set, the horizon, and the prediction model (7). All the deployed values are those of the profile `planner_params_g1.yaml` shipped with the code.

Unitree G1	
Morphology	bipedal humanoid
Actuated joints	29 (15 driven by the locomotion policy)
Exteroception	Livox MID-360 LiDAR, 8 m usable range
Locomotion layer	AMO reinforcement-learning policy
Layer update rate	50 Hz
Command interface	body-frame twist u , remapped (Section 2.3.1)
Admissible input set	$v_x \in [-0.20, 0.40]$ m/s, $ v_y \leq 0.20$ m/s, $ \omega \leq 0.40$ rad/s
Horizon	$N = 15$, $\Delta t = 0.35$ s (5.25 s of preview), re-solved at 8 Hz
Abstraction used by the MPC	first-order velocity lag, $\tau_v = \tau_w = 10^{-3}$ s (degenerate, Section 2.5)

evasive manoeuvres the machine can never execute and would pass closer to obstacles than the clearance radius promises. In translation the deployed bound keeps a margin — 0.40 against 0.5 m/s — but in rotation it does not: 0.40 rad/s *is* the trained ceiling, so that axis runs with no margin at all, and any degradation of the policy near its limit reaches the optimizer as unmodelled lag rather than as a violated bound. The lateral bound is smaller than the forward one for a reason that is kinematic and not conservative: a biped strafes genuinely more slowly than it walks forward.

2.3. Low-level locomotion layer

The controller analysed in this report never commands a joint. Its decision variable is a body-frame twist, and between that twist and the ground sits a layer that turns it into joint commands at 50 Hz. Only two properties of that layer reach the optimization problem: the twist is tracked with a first-order lag rather than instantaneously, and the tracking is fast enough, relative to the 8 Hz MPC cycle, for the two layers to be designed separately. This section describes that locomotion controller to the depth needed to justify those two properties and no further. Throughout, $q \in \mathbb{R}^{n_j}$ denotes the vector of joint positions ($n_j = 29$), $\mathcal{T} \in \mathbb{R}^{n_j}$ the vector of joint torques, and $K_P, K_D \succ 0$ the diagonal proportional and derivative gain matrices of the joint-level loop; the symbol τ is reserved for the actuator time constants of Section 2.5.

2.3.1 Learned whole-body control: the AMO policy

On a humanoid the body-twist interface is produced by a neural network rather than by a hand-designed gait generator. The deployed controller is AMO (Adaptive Motion Optimization) [4], a reinforcement-learning (RL) policy trained in simulation for the G1 and its twenty-nine degrees of freedom (DoF). AMO is deliberately *not* a walking controller: it is a loco-manipulation policy, trained to track torso height, torso orientation and upper-body targets while walking, on a hybrid dataset in which an offline trajectory-optimization module supplies whole-body references that are kinematically consistent with the commanded end-effector motion. The navigation stack exercises only its locomotion subset, the arms are held at the default posture and only the base command is driven, but the policy is the one trained for the larger task, which is what makes the same platform usable for manipulation later without changing the layer the MPC talks to.

Structure. At each tick k of the 50 Hz policy loop, the network maps an observation to an action,

$$a_k = \pi(o_k) \in \mathbb{R}^{15}, \quad o_k = (\omega_k^b, \hat{g}_k, \tilde{u}_k, q_k - q^{\text{def}}, \dot{q}_k, a_{k-1}), \quad (1)$$

where $\omega_k^b \in \mathbb{R}^3$ is the body angular velocity, $\hat{g}_k \in \mathbb{R}^3$ the gravity direction expressed in the body frame (the tilt an inertial measurement unit (IMU) provides without a global heading), $q_k, \dot{q}_k$ the measured joint positions and velocities of the 23 observed joints, q^{def} the default posture, a_{k-1} the previous action, and $\tilde{u}_k$ the command. The action is not a torque: it is an offset on the default posture, which a fixed joint-level proportional–derivative (PD) law converts into torque,

$$q_k^{\text{ref}} = q^{\text{def}} + s_a a_k, \quad \mathcal{T}_k = K_P(q_k^{\text{ref}} - q_k) + K_D(0 - \dot{q}_k), \quad (2)$$

with s_a a fixed action scale and a zero reference joint velocity. The learned part of the controller is therefore exactly the map from observation to PD setpoint; the stabilising loop underneath it is a fixed impedance, untouched by learning, and it is what lets the leg absorb terrain irregularity and impact by programmable compliance rather than by contact sensing or force control. The policy π is obtained by proximal policy optimization (PPO) [9], maximising the expected discounted return $\mathbb{E}[\sum_k \gamma^k \mathcal{R}_k]$ with $\gamma \in (0, 1)$ and a stage reward $\mathcal{R}_k$ that pays for tracking the commanded twist and penalises posture deviation, joint effort and action rate; masses, friction, gains and latencies are randomised during training so that the fixed policy transfers to hardware [8].

The command mapping, and one unmodelled nonlinearity. The command handed to the network is not the MPC twist but its image under a calibrated map, $\tilde{u}_k = \mathcal{M}(u_k)$, and that map is not a change of units: the policy’s yaw channel is a heading *offset* rather than a yaw rate, its lateral channel uses the opposite sign convention, and the policy plants its feet whenever all three channels fall below a stand gate. That gate is the one visible unmodelled nonlinearity of the cascade — a near-pure rotation in place, which the MPC produces when the reference turns sharply at low speed, falls below it and returns no motion at all. It is a discontinuity in the map from the optimizer’s output to the robot’s motion, and it is not represented anywhere in the program.

2.3.2 The cascade and its abstraction seam

Fig. 2 places the locomotion layer in the control structure it belongs to. The stack is a cascade of four loops whose rates are separated by roughly a decade each: the A^* reference is recomputed at 1 Hz, the MPC at 8 Hz, the locomotion policy at 50 Hz, and the joint PD loop at the motor-driver rate. That separation is the standard justification for designing each layer against a static abstraction of the one below it, and the abstraction used here is deliberately minimal: three decoupled first-order lags, Eq. (7), whose constants are discussed in Section 2.5. A trained network, a fixed impedance loop and two legs are represented, inside the optimization program, by two numbers.

It is worth stating what that abstraction discards, because it is the seam at which the whole design is exposed. It discards leg dynamics, contact, footstep timing, the derating the policy applies to its own command, and the stand gate of Section 2.3.1. What survives is a question of degree rather than of principle, and it is measured, not assumed, in Section 5.3.

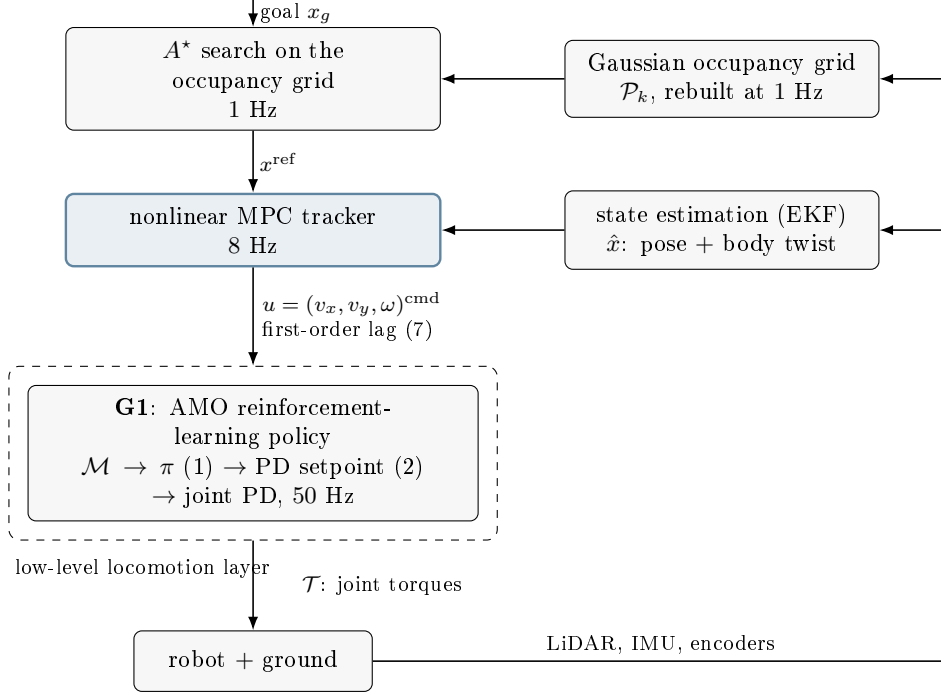


Figure 2: Cascaded structure of the controller, from the goal down to joint torques. The rates fall by roughly a decade at every seam (1 Hz, 8 Hz, 50 Hz, motor-driver rate), which is what allows the layers to be designed separately. Everything inside the dashed box is invisible to the optimization program, which sees only the twist interface u and the first-order lag by which it is tracked.

2.4. Environment representation

Let $\mathcal{P}_k = \{p_i \in \mathbb{R}^3\}_{i=1}^{N_k}$ be the LiDAR point cloud available at cycle k , projected on the plane. The grid is a square of side $2h_w$ ($h_w = 6$ m) centred on the robot position, discretised with resolution $\Delta = 0.20$ m into $n_c = 2h_w/\Delta = 60$ cells per axis. The occupancy probability of cell c is a Gaussian tail of the distance to the nearest return,

$$P(c) = 1 - \Phi\left(\frac{d_{\min}(c, \mathcal{P}_k)}{\sigma}\right), \quad d_{\min}(c, \mathcal{P}_k) = \min_{p \in \mathcal{P}_k} \|c - p\|_2, \quad (3)$$

with Φ the standard normal cumulative distribution function,

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-t^2/2} dt = \frac{1}{2} \left(1 + \operatorname{erf}\left(\frac{x}{\sqrt{2}}\right)\right) \in (0, 1), \quad (4)$$

and $\sigma = 0.31$ m. Equation (3) is an *inflation* model: it is not a Bayesian occupancy update (there is no recursive filtering, no ray casting and no free-space carving), but a smooth, monotone map from clearance to cost, whose one parameter σ sets the width of the obstacle halo. Cells with $P(c) \geq P_{\text{thr}} = 0.10$ are hard obstacles.

The pair (σ, P_{thr}) is not chosen by hand and then justified: inverting (3) gives the hard-blocking radius that a threshold implies,

$$d_{\text{block}} = \sigma \Phi^{-1}(1 - P_{\text{thr}}), \quad (5)$$

and the deployed pair is the one for which $d_{\text{block}} = 0.31 \times 1.2816 = 0.40$ m against a body radius of about 0.35 m. The parameter that has a physical meaning is the clearance; the two grid parameters are derived from it. The left panel of Fig. 3 shows the trade-off σ governs — a small σ leaves doorways open but hugs walls, a large one closes gaps the robot could physically cross — with the deployed curve and the resulting d_{block} marked.

Because the grid is rebuilt from scratch, its construction cost is the dominant cost of the planning layer, $O(n_c^2 |\mathcal{P}_k|)$; vectorised over 60×60 cells it measures a few milliseconds together with the A^* search, which at the 1 Hz replanning rate is negligible against the 8 Hz MPC.

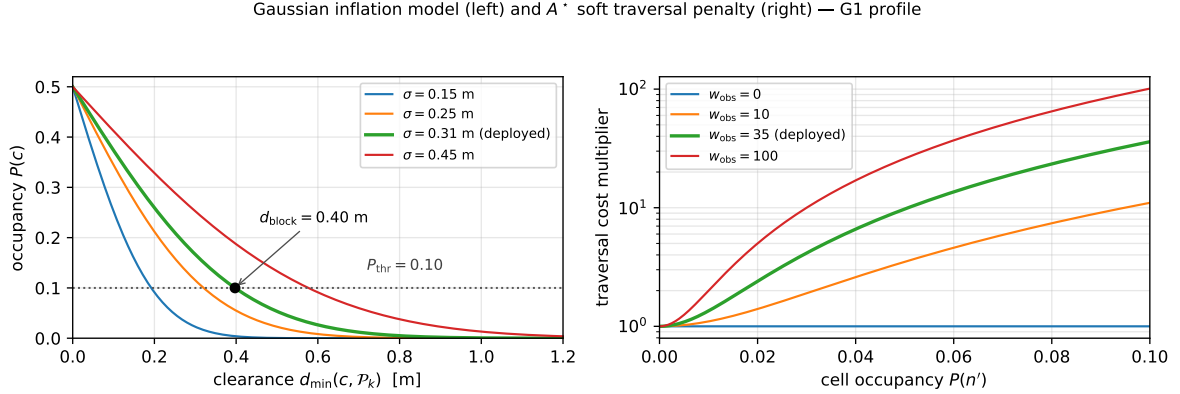


Figure 3: Left: the Gaussian inflation model (3) for four values of σ , with the deployed curve in bold, the hard threshold $P_{\text{thr}} = 0.10$ and the blocking radius d_{block} it implies through (5). Right: the soft traversal-cost multiplier $1 + w_{\text{obs}}(P/P_{\text{thr}})^2$ of the A^* edge cost (10), which is what actually keeps the path away from walls; the deployed value is $w_{\text{obs}} = 35$. Both panels are drawn from the deployed parameter file rather than transcribed from it.

2.5. Plant model and state-space formulation

The MPC model must predict where the robot will be, not where it was told to go. Commanding a legged platform a body-frame velocity does not, in general, produce that velocity instantaneously: the locomotion controller responds with a time constant of the order of a tenth of a second, which over a multi-second horizon accumulates into a prediction error of several tens of centimetres if neglected. The model therefore augments the usual planar kinematic state with the three velocity states, giving

$$x = [p_x \ p_y \ \psi \ v_x \ v_y \ \omega]^\top \in \mathbb{R}^6, \quad u = [v_x^{\text{cmd}} \ v_y^{\text{cmd}} \ \omega^{\text{cmd}}]^\top \in \mathbb{R}^3, \quad (6)$$

where (p_x, p_y) and ψ are the world-frame pose and (v_x, v_y, ω) the body-frame twist actually realised by the platform. In continuous time the actuator behaves as three decoupled first-order lags driven by the commands,

$$\dot{v}_x = \frac{u_1 - v_x}{\tau_v}, \quad \dot{v}_y = \frac{u_2 - v_y}{\tau_w}, \quad \dot{\omega} = \frac{u_3 - \omega}{\tau_w}, \quad (7)$$

while the pose obeys the holonomic body-to-world rotation¹

$$\dot{p}_x = v_x \cos \psi - v_y \sin \psi, \quad \dot{p}_y = v_x \sin \psi + v_y \cos \psi, \quad \dot{\psi} = \omega. \quad (8)$$

Model (7)–(8) is holonomic: unlike a differential-drive vehicle, the platform can command a lateral velocity, so no nonholonomic constraint appears and the input set is a box. This is what makes the resulting NLP comparatively benign, the nonlinearity is confined to the rotation $R(\psi)$ and to the obstacle distances.

¹In the deployed code the lateral velocity v_y is propagated with τ_w rather than with τ_v , i.e. lateral motion is assumed to respond as fast as yaw rather than as fast as forward motion. Under the present profile the two constants are equal and the choice is numerically immaterial; it is reported because the analysis reproduces the deployed formulation exactly rather than an idealisation of it.

The deployed time constant, and why it is degenerate. The G1 profile sets $\tau_v = \tau_w = 10^{-3}$ s. This is not an identified value and is not meant to be one. The plant used throughout this report integrates the commanded twist *kinematically*, with no actuation dynamics at all, so the zero-order-hold coefficient of Section 3.2 evaluates to $1 - e^{-\Delta t/\tau} = 1$ in double precision and the velocity update collapses to $v_{k+1} = u_k$: the three velocity states are not dynamic states but delayed copies of the input. The choice is deliberate and its purpose is methodological — with the prediction model equal to the simulation model by construction, what the experiments measure is the behaviour of the *optimizer* and not the noise of a gait. Its cost is equally explicit: any conclusion that depends on the actuator pole, the selection of the sampling time from the plant bandwidth being the obvious one, cannot be drawn from these data and must wait until τ is identified under physics and the AMO policy. The lag structure is nevertheless kept in the model, because it is the term that will carry the identified value the moment there is one, and because removing it would change the dimension of the program and invalidate every comparison in this report.

2.6. Navigation problem statement

Let $x_g \in \mathbb{R}^2$ be the goal. The navigation task is the receding-horizon optimal control problem

$$\min_{u_0, \dots, u_{N-1}} J(x_{0:N}, u_{0:N-1}, \mathcal{P}_k, \theta) \quad (9a)$$

$$\text{s.t. } x_{t+1} = f(x_t, u_t), \quad t = 0, \dots, N-1, \quad (9b)$$

$$x_0 = \hat{x}(t_k), \quad (9c)$$

$$u_t \in \mathcal{U}, \quad t = 0, \dots, N-1, \quad (9d)$$

$$d(x_t, \mathcal{P}_k) \geq r_{\min}, \quad t = 0, \dots, N, \quad (9e)$$

solved at every control instant t_k . Two properties of the way (9) meets the robot are worth stating before the design is read, because both are deviations from the textbook receding-horizon implementation.

The sampling time of the model is not the period of the loop. The step Δt in (9b) is the *prediction* step, 0.35 s in the deployed profile, while the program itself is re-solved at 8 Hz. The horizon is therefore re-planned about 2.8 times per predicted node, and the first predicted node already lies 0.35 s of open loop ahead of the state that produced it. The two quantities are independent design variables and are treated as such throughout.

Only u_0 would be applied — but it is not. The tracker does not publish u_0^* . It walks the predicted trajectory $x_{0:N}^*$ out to a fixed look-ahead distance and publishes the point it reaches as a geometric setpoint; a proportional law downstream converts that setpoint into the body twist actually sent to the locomotion layer, at 20 Hz. The optimizer is thus used as a *reference generator* rather than as a controller, and the loop closed on the robot is the cascade of that law with the locomotion policy rather than (9) itself. Section 5.1 bypasses the stage deliberately, so that the numerical results measure the optimization layer and not the tuning of the law beneath it.

Three features of (9) then determine the whole design:

- **The goal is not reachable within the horizon.** With $N\Delta t = 5.25$ s and a cruise speed of 0.27 m/s the horizon spans about a metre and a half, while the goal may be tens of metres away and outside the sensed region altogether. The cost (9a) therefore cannot be a distance to x_g ; it must be a distance to a *reference trajectory* produced by a separate, cheaper program that looks further ahead. That program is the A^* search of Section 3.1.
- **The obstacle set is a point cloud, not a polytope.** Constraint (9e) is non-convex, and its description changes at every cycle. It is relaxed into a smooth penalty (Section 3.3.2), which keeps the NLP smooth and always feasible at the price of losing the hard safety guarantee.
- **The cost is parametrised by θ .** The eleven weights in θ are not physical quantities; they are design variables of a second-level optimization problem (Section 5.8).

3. Formulation of the optimization program

3.1. Reference generation: A^* as a discrete program

Section 2.6 established that the goal lies far outside the horizon, so the cost of (9a) must measure the distance to a reference trajectory rather than to x_g . That reference is produced once per replanning cycle by a discrete search on the occupancy grid of Section 2.4. Cells above the occupancy threshold are removed from the graph;

the remaining 8-connected edges carry the cost

$$g(n \rightarrow n') = c_{\text{move}} \cdot \delta \cdot \left[1 + w_{\text{obs}} \left(\frac{P(n')}{P_{\text{thr}}} \right)^2 \right], \quad c_{\text{move}} \in \{1, \sqrt{2}\}, \quad (10)$$

and a standard A^* search returns the cheapest path between two cells. The normalised quadratic in (10) is what produces medial-axis-like paths: a cell just below the hard threshold costs $(1 + w_{\text{obs}})$ times an open cell, so the search prefers the middle of a corridor to a shorter path that grazes a wall, while a doorway is never closed outright. The Euclidean heuristic is admissible because the bracket is bounded below by 1, so no edge can be cheaper than its Euclidean length; the search is therefore an exact A^* and returns a cost-optimal path on the current occupancy grid.

What the discrete layer is for. Choosing *which side* of an obstacle to pass on is a discrete decision. Take a trajectory that passes on the left and one that passes on the right: turning the first into the second means moving it continuously through intermediate trajectories, and at least one of those must run through the obstacle. The feasible set is therefore disconnected, with one component per side, and a gradient-based solver stays in the component its initial guess falls into: it converges to the best trajectory on one side and never learns that the other side existed. Writing the choice into the NLP would require binary variables and turn it into a mixed-integer nonlinear program, whose solution time is not compatible with a control loop at 8 Hz. A graph search represents the same choice natively, and on a 60×60 grid it costs a few milliseconds. Delegating the decision to the A^* algorithm is therefore what leaves the continuous program smooth, with a box-shaped input set and no integer variables — the structure that Sections 4.1 and 4.3 exploit throughout.

The price to pay for such delegation is that the choice is made before the MPC is consulted and the MPC has no way to revise it: it can only deform the trajectory within the class it has been given. The two ways this breaks are opposite to each other and both are measured in Section 5. When the discrete layer *does not commit* — an obstacle squarely on the line to the goal, two routes of equal cost — the smooth program inherits two competing basins of its own, and small perturbations flip the solution from one to the other (Section 5.4). When it *commits on poor information*, the MPC follows it faithfully into a dead end, because questioning the reference is not its job (Section 5.9.1).

Choosing the target inside the window. The search of (10) runs on the local grid, and the global goal is normally outside it, so a target on the window has to be selected first. This is a selection over the free cells of the window boundary,

$$x_{\text{local}} = \arg \min_{c \in \partial \mathcal{W}(\hat{x}) \cap \mathcal{F}_k} \left[d_k(c, x_g) + w_\tau \tau_k(c) \right], \quad (11)$$

where $\mathcal{F}_k$ are the free boundary cells, d_k is a distance to the goal, and τ_k penalises cells the robot has already visited without making progress. Only boundary cells are candidates, because the boundary is where the direction of travel is decided; an interior cell would stop the robot half a window short of the edge for no reason. The substantive choice in (11) is the metric d_k . Two target rules were implemented and compared; only the second is an instance of (11) at all, and that is itself part of the story.

The first, which the stack shipped with, does not solve (11) at all. It takes the last free cell along the ray from the robot to the goal, which is a geometric shortcut rather than a minimisation,

$$x_{\text{local}} = \begin{cases} \text{cell}(x_g), & x_g \in \mathcal{W}(\hat{x}), \\ \text{ray}(\hat{x} \rightarrow x_g) \cap \partial \mathcal{W}(\hat{x}), & \text{otherwise,} \end{cases} \quad (12)$$

and the robot advances one grid width at a time towards the goal; if the projected cell is occupied, a breadth-first search returns the nearest free one. The rule is cheap, needs no map at all, and is memoryless and purely directional: whatever the map contains, the target lies on the line to the goal. In front of a concave obstacle that line enters the concavity, so the target does too.

It would be convenient if the defect stopped there, because then (11) with a Euclidean d_k would already repair it. It does not. Figure 4 shows the robot at the mouth of a horseshoe with the map it has actually observed: the ray projection selects a boundary cell 30.2 m of travel from the goal, and the $\arg \min$ of $\|c - x_g\|_2$ over the same free boundary cells selects one 30.4 m away — a different cell, and just as wrong. Both are near the closed end, because in a straight line both are close to the goal. The straight-line distance rates a cell inside the concavity and one outside it almost equally, and where the two happen to tie exactly it separates them on numerical noise and reverses the choice at the next cycle. Section 5.9.1 shows that this is enough, on its own, to trap the robot.

The second instance is the deployed one. It keeps (11) and replaces d_k with the *geodesic* distance to the goal over the free cells of the accumulated map: a wavefront propagated from x_g , one scalar field per replanning cycle, with no path extracted from it. Space that has not been observed yet counts as free. This is the optimistic assumption of frontier exploration, and it is what makes the field correct itself as the sensor discovers: while the far end of a corridor is unseen the wavefront runs through it and entering is the right decision, and as soon as that end is observed the same cell becomes distant and the target is dropped. A candidate the wavefront cannot reach at all is discarded rather than scored with a Euclidean fallback, which would readmit the metric at fault; if no candidate is reachable on the known map the planner falls back to (12), which at least moves the robot towards unknown ground. The memory term $w_\tau \tau_k$ is implemented but carries zero weight in the deployed profile, for the reason measured in Section 5.9.2. In the same scene the geodesic argmin selects a cell 7.4 m from the goal, on the far side of the obstacle’s arm, and A^* routes out of the concavity instead of further into it.

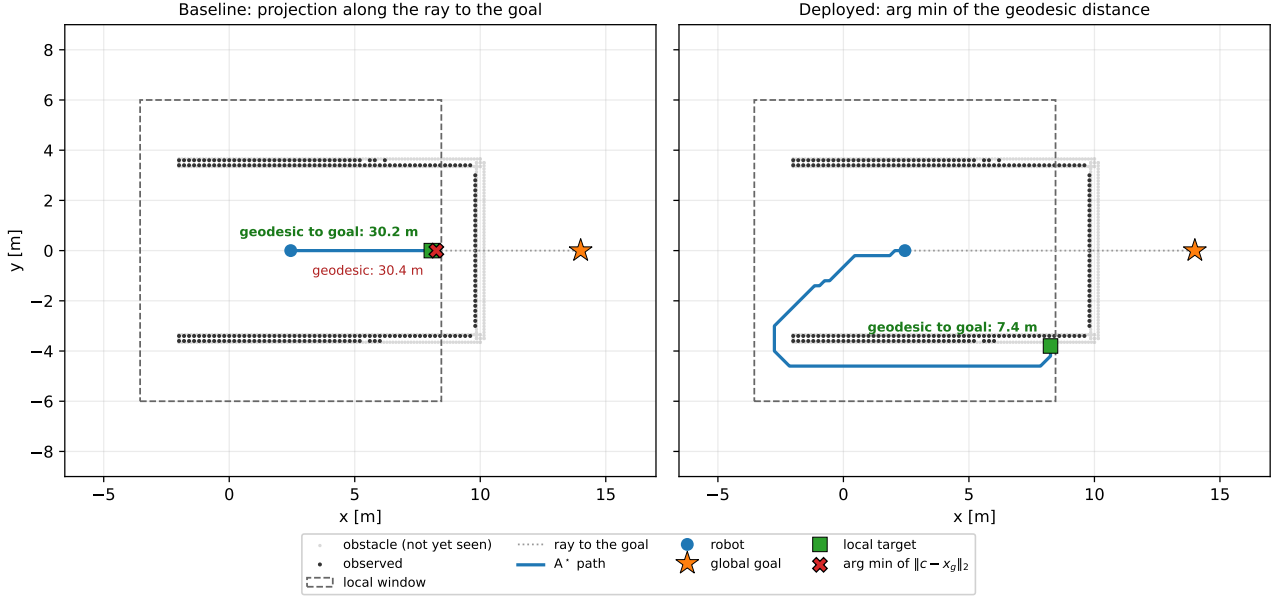


Figure 4: The same scene, the same window and the same observed map; only the metric of (11) changes. The robot sits at the mouth of a horseshoe and the map is the one it has actually accumulated by driving in, not the full world (light grey), which is what makes the failure reproducible. Left: the deployed projection (12) puts the target at the closed end, and the argmin of the Euclidean distance over the same free boundary cells lands beside it — the two rules differ, the outcome does not, because both are ~ 30 m of travel from the goal. Right: with the geodesic d_k the target moves to the far side of the lower arm, 7.4 m away, and the A^* solution leaves the concavity. Drawn from the deployed planner and profile, not sketched.

Commitment between replanning cycles. Two boundary cells lying on opposite sides of an obstacle can carry almost the same geodesic cost while belonging to different homotopy classes. The argmin then alternates between them from one cycle to the next, and the robot sways on the spot without ever committing. The deployed rule therefore retains the candidate that best continues the previous choice unless a competitor beats it by a margin of 2.0 m of geodesic cost, and clears that memory whenever the global goal changes, since a target inherited from a previous mission is simply the wrong direction. This is the ambiguity of Section 5.4 seen one layer higher up: where the smooth program resolves it by breaking symmetry numerically, the discrete layer resolves it by declining to switch without evidence.

The geodesic field is the one expensive part of the planning layer, because the wavefront covers the whole known map rather than the window alone: it measures a median of 0.11 to 0.15 s per replanning cycle depending on the world, with a worst case of 0.27 s, which against the 1 Hz replanning rate is between a tenth and a quarter of the period and leaves the 8 Hz MPC untouched. The resulting grid path is finally resampled and smoothed with a moving average before it is handed to the MPC, to remove the zig-zag that the 8-connected discretisation introduces.

3.2. Model discretization

With sampling time Δt , the lag subsystem (7) is linear and time-invariant, so it is discretised *exactly* under a zero-order-hold input. Defining $\nu_v = 1 - e^{-\Delta t/\tau_v}$ and $\nu_w = 1 - e^{-\Delta t/\tau_w}$, the velocity update is the exact solution of (7) over one step,

$$v_{x,k+1} = (1 - \nu_v)v_{x,k} + \nu_v u_{1,k}, \quad \omega_{k+1} = (1 - \nu_w)\omega_k + \nu_w u_{3,k}, \quad (13)$$

and analogously for v_y . No approximation is involved here and none should be claimed: this is the Lagrange formula for a linear time-invariant (LTI) system under a piecewise-constant input, and it is exact at the sampling instants. On the deployed G1 profile it is straightforward, since $\nu = 1$ reduces (13) to $v_{k+1} = u_k$; the statement is retained because it is the part of the model that would carry the weight on hardware, where τ is real.

The pose, by contrast, is genuinely nonlinear and must be integrated numerically. Two possibilities are implemented and compared:

$$\text{explicit Euler: } p_{k+1} = p_k + R(\psi_k) v_{k+1} \Delta t, \quad (14)$$

$$\text{mid-point (RK2): } p_{k+1} = p_k + R(\psi_k + \tfrac{1}{2}\omega_{k+1}\Delta t) v_{k+1} \Delta t. \quad (15)$$

Both evaluate the rotation on the *post-lag* velocity v_{k+1} , a semi-implicit choice that costs nothing and removes the one-step delay that an explicit scheme would introduce between a command and the predicted displacement. The mid-point rule differs from Euler by one addition inside the argument of the rotation.

The results shown in the following compare two integrators outside the optimizer, which is only meaningful if the map assembled inside the NLP is the same one. That is checked rather than assumed: a single transition is extracted from the CasADi program built by the tracker, by reconstructing $f(x_0, u_0)$ from the first six equality constraints, and compared with the reference implementation of (14) and (15). The residual is zero to machine precision for both schemes.

Errors quantification according to the discretization method. The global error of each scheme was first fitted against the step size on three motion regimes, over a 2.80 s window, with the exact solution available in closed form: at constant body velocity the pose traces a circular arc. The measured orders are 1.00 and 2.00, exactly the first and second order the truncation analysis predicts, and the result is identical across regimes. The step grid is built by halving from the deployed Δt , so the fit begins at the step the controller actually uses rather than near it.

It is worth noting that a constant-velocity arc is not the regime in which the MPC works. The optimizer applies a *different* input at every node, and over a real horizon the commanded yaw rate changes sign.

The comparison is repeated on 12 horizons taken from a bag of data recorded while G1 is navigating in a mapless MuJoCo world and re-solved with the deployed profile, each integrated with its own optimal input sequence. The step is varied by subdividing the prediction interval while the input stays constant over each Δt , which keeps an order measurable on a real trajectory; the reference is the exact arc on the same post-lag velocity sequence, so the only difference between the three trajectories is how $R(\psi)$ is evaluated.

Table 2: Median position error at the end of the horizon, over 12 horizons re-solved from a bag of data recorded during the navigation, against the exact arc on the same velocity sequence. The first row is the deployed step; the others subdivide the prediction interval without changing the input.

Δt [s]	error, Euler [m]	error, mid-point [m]	ratio
0.3500	2.08×10^{-2}	3.66×10^{-4}	57
0.1750	1.04×10^{-2}	9.14×10^{-5}	113
0.0875	5.18×10^{-3}	2.28×10^{-5}	227
0.0437	2.59×10^{-3}	5.71×10^{-6}	453
0.0219	1.29×10^{-3}	1.43×10^{-6}	906

The orders survive the change of regime — 1.00 and 2.00 — but the error does not. At the deployed step the median is 2.08×10^{-2} m for Euler against 3.66×10^{-4} m for the mid-point rule, a factor 57 where the synthetic arc reports 114; the tails are 3.69×10^{-2} m and 7.16×10^{-4} m. The gap is the regime and not an inconsistency: over these horizons the commanded yaw rate spans 0.54 rad/s, so the per-step errors partially cancel each other out, while the arc accumulates them along a single turn. Measuring on the synthetic regime alone would have overestimated the benefit by a factor of two.

Where the floor is. One number bounds the discussion. Holding the velocity at its post-lag value across the interval — the semi-implicit choice made in (14) and (15) — neglects a displacement of 8.01×10^{-4} m per horizon, because the model’s velocity rises towards the command instead of arriving at it. That is larger than the mid-point truncation error itself. Below that level refining the pose scheme doesn’t improve the results: a different modelling choice already dominates and the second order is not the binding approximation.

Closed loop analysis. Prediction fidelity and executed cost are different quantities. Running the two schemes on the same worlds with everything else fixed moves the median closed-loop cost by at most 2.3%, and the sign changes from world to world — the signature of run-to-run variation rather than of an improvement.

Table 3: Closed-loop cost with the two integrators, same profile otherwise. A positive Δ means the mid-point rule is cheaper.

world	median cost, Euler	median cost, mid-point	Δ [%]
u_trap	163.5	167.2	-2.3
narrow_gap	110.7	108.2	2.3

The reason is structural and was stated in Section 2.6: only the first input is ever applied, and A^* replans at 1 Hz, so the accuracy of the far end of the prediction reaches the executed trajectory only indirectly. The mid-point rule is deployed because it is a strictly better prediction for the price of one addition, not because it was measured to steer better.

Section 5.3 confirms the thesis: measured against the plant, the prediction diverges by 38 times the Euler error over the same horizon and 2180 times the mid-point error. The integrator was never the dominant term.

3.3. The nonlinear MPC program

3.3.1 Decision variables, cost, and the parametrisation of the program

The NLP is written in the *simultaneous* form, also called direct multiple shooting: both the state and the input sequences are decision variables and the dynamics enter as equality constraints,

$$z = (X, U), \quad X = [x_0, \dots, x_N] \in \mathbb{R}^{6 \times (N+1)}, \quad U = [u_0, \dots, u_{N-1}] \in \mathbb{R}^{3 \times N}. \quad (16)$$

With $e_k = x_k - x_{\text{ref},k}$ the cost is

$$J(z; \theta) = \sum_{k=0}^{N-1} \left[e_k^\top Q e_k + u_k^\top R u_k + J_{\text{obs}}(x_k) \right] + \sum_{k=1}^{N-1} R_{\text{jerk}} \|u_k - u_{k-1}\|_2^2 + e_N^\top Q_T e_N + J_{\text{obs}}(x_N), \quad (17)$$

with $Q = \text{diag}(Q_x, Q_y, Q_\psi, 0, 0, 0)$, $Q_T = \rho_T Q$ and $R = \text{diag}(R_{v_x}, R_{v_y}, R_\omega)$. Three properties of (17) deserve comment.

The stage weight is only semi-definite. No velocity state is penalised: Q has rank 3 out of 6. The velocities are shaped indirectly, through the lag dynamics and through the input penalty R . This is what keeps the reference generator simple, since an A^* path carries no velocity information, and the price is that the cost contains no direct damping of the velocity states.

The rate penalty couples consecutive inputs. The term $R_{\text{jerk}} \|u_k - u_{k-1}\|_2^2$ is the discrete analogue of an input-rate penalty, and it is the only term that puts off-diagonal blocks between adjacent input stages in the Hessian. Its purpose is physical: a legged platform tracks a smooth velocity profile far better than a chattering one. Note that the sum starts at $k = 1$: the transition from the command applied on the previous cycle to u_0 is not penalised, so the first predicted move is free to jump. That is a deliberate omission — the program is re-solved at 8 Hz and would otherwise inherit a memory of its own past output — but it means the smoothness the term buys is smoothness *within* a horizon, not across consecutive ones.

The terminal cost is a scaled copy of the stage cost. $Q_T = \rho_T Q$ is a heuristic, neither the solution of a Riccati equation nor an exact cost-to-go, and like Q it acts only on the three position and heading components. ρ_T is therefore a tuning parameter and is treated as one, appearing among the eleven weights of θ in Section 5.8. (The gap between this heuristic and the true infinite-horizon cost-to-go of the linearised dynamics is not quantified in this report; doing so would require the Riccati comparison, which the semi-definite Q would first oblige us to regularise.) The stability argument is not carried by Q_T at all, but by the terminal equilibrium constraint of Section 3.3.5.

Comparison between multiple and single shooting. The standard argument for multiple shooting is that it keeps the Jacobian sparse and banded, where a condensed single-shooting formulation, in which states are eliminated by recursive substitution and so the inputs are the only variables, would produce a small but dense problem. A direct comparison between the two approaches can be performed, with the aim to show why a multiple-shooting approach was chosen.

Table 4: Multiple (M) against single (S) shooting on the same instance, at the deployed profile. Dimensions, densities and iteration counts are exact; each timing is the fastest of three cold solves, with no warm start. The table shows the comparison for the number of variables, number of constraints, the Hessian density, the number of iterations and the solve time needed by each method for different lengths of the prediction horizon N .

N	vars M / S	cons M / S	hess dens. [%] M / S	iter M / S	solve [ms] M / S
5	51 / 15	56 / 20	12.11/100.00	19 / 18	71 / 64
10	96 / 30	106 / 40	6.51/100.00	29 / 21	137 / 135
15	141 / 45	156 / 60	4.45/100.00	35 / 24	188 / 218
25	231 / 75	256 / 100	2.73/100.00	39 / 31	299 / 524
40	366 / 120	406 / 160	1.72/100.00	39 / 39	393 / 961
60	546 / 180	606 / 240	1.16/100.00	39 / 29	593 / 2106

The structural half of the claim holds exactly, and the Hessian is where it shows: at $N = 15$ the program shrinks from 141 variables and 156 constraints to 45 and 60, while the Hessian density goes from 4.45% to 100.00%, a full matrix. Condensing does trade a large banded problem for a small dense one.

If the last two columns together are analyzed, the reason why employing a multiple-shooting approach is beneficial becomes clear. At $N = 60$ the condensed form converges in *fewer* iterations than the sparse one, 29 against 39, and still takes three and a half times as long. The cost is therefore not in the number of Newton steps but in what each one costs, which is a factorisation of the Hessian, that as shown in the middle column goes from a few per cent dense to a full dense matrix.

Two caveats keep this from being a general result. First, at $N = 15$ and $N = 40$ the two parametrisations converge to *different* minima, 6.7% and 5.8% apart in cost. That is not an implementation error — the program is non-convex and two parametrisations follow different optimization paths, so they can land in different basins — but it does mean that on those two rows, one of which is the deployed horizon, the timings compare solves that did not solve the same problem. Second, the conditioning argument usually made for multiple shooting does not show up here at all: single shooting integrates the model in *open loop* over the whole horizon, so its error compounds step by step and the problem degrades with N and with any instability of the plant, but on a kinematic and stable model that defect never surfaces. This comparison therefore supports the choice made here; it is not evidence for the general case.

3.3.2 Obstacle avoidance as a smooth penalty

The clearance constraint (9e) is replaced, in the deployed configuration, by a penalty evaluated on the $M = 8$ nearest LiDAR returns $\{p_j\}$ inside a search radius of 2.0 m:

$$J_{\text{obs}}(x_k) = W_{\text{obs}} \sum_{j=1}^M \left[\underbrace{\frac{1}{2} \left(1 - \tanh \left(\frac{\alpha_{\text{obs}}}{2} (d_{k,j} - r_{\text{obs}}) \right) \right)}_{\text{sigmoid zone}} + \underbrace{2 \max(0, r_{\text{obs}} - d_{k,j})^2}_{\text{penetration penalty}} \right], \quad (18)$$

with $d_{k,j} = \sqrt{(p_{x,k} - p_{j,x})^2 + (p_{y,k} - p_{j,y})^2 + \varepsilon_d}$ and $\varepsilon_d = 10^{-6}$ making the distance differentiable at zero, where the Euclidean norm is not. The sigmoid is written with $\tanh$ rather than as $1/(1 + e^{\alpha_{\text{obs}}(d - r_{\text{obs}})})$: the two are algebraically identical, but the logistic form overflows for large arguments while $\tanh$ saturates cleanly. The argument does get large: the sentinel padding described below places absent returns at 10^3 m, which at $\alpha_{\text{obs}} = 6.00$ makes the exponent $\approx 6 \times 10^3$ — far past the point where e^x overflows a double, while $\tanh$ simply returns 1.

The hybrid structure is deliberate, and Fig. 5 is the argument for it. The sigmoid saturates at W_{obs} as $d \rightarrow 0$: on its own it is a *bounded* penalty, so a sufficiently strong tracking term can always buy its way through an obstacle. Worse for the solver, its gradient decays on both sides of r_{obs} , so the restoring force fades exactly where the robot most needs pushing out. At the deployed $\alpha_{\text{obs}} = 6.00$ the decay is mild — at two centimetres of clearance the sigmoid still retains about a third of the restoring force it has at $d = r_{\text{obs}}$ — but it is governed

by α_{obs} and it sharpens fast: at four times that steepness the same point retains essentially none. Steepening the barrier to improve avoidance is therefore precisely what destroys its gradient where avoidance has already failed. The quadratic term removes both defects inside the safety radius: it is unbounded, and its gradient grows linearly with penetration regardless of α_{obs} .

The price is the third panel. The curvature of (18) scales as $\alpha_{\text{obs}}^2 W_{\text{obs}}$, so a steep, heavy penalty injects large and rapidly varying eigenvalues into the Hessian precisely where the solution lives. The deployed tuning sits at the crossover: the peak curvature contributed by the sigmoid, $\approx 0.096 \alpha_{\text{obs}}^2 W_{\text{obs}}$, is just below the $4W_{\text{obs}}$ contributed by the quadratic term, so neither dominates. Doubling α_{obs} puts the sigmoid a factor of three above it, quadrupling it a factor of fourteen, and the conditioning of the NLP follows.

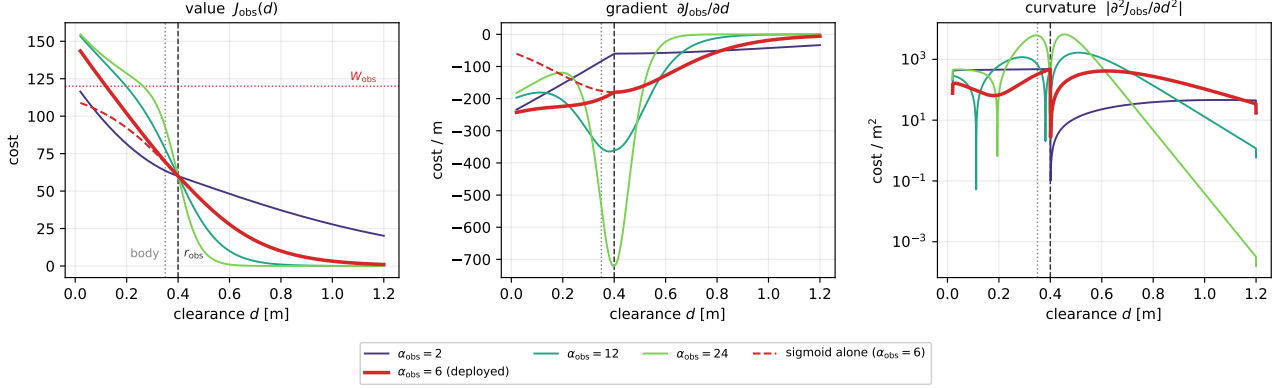


Figure 5: The hybrid penalty (18) against clearance, for four steepness values with the deployed $\alpha_{\text{obs}} = 6.00$ in bold ($W_{\text{obs}} = 120$, $r_{\text{obs}} = 0.40$ m; the dotted line marks the body radius). Dashed: the sigmoid zone alone at the deployed steepness, which saturates at W_{obs} and whose gradient fades inside the obstacle — the two defects the quadratic term repairs. Right, on a logarithmic scale: the curvature, which grows as α_{obs}^2 and is what degrades the conditioning of the NLP; the notches are sign changes. Evaluated through the same function the tracker builds into the NLP, not redrawn from the formula.

Relaxing a hard constraint into a penalty has three consequences worth stating explicitly. (i) The NLP is always feasible: no state constraint can render the problem unsolvable when the robot is already too close to an obstacle, which for a real-time loop with no fallback trajectory is a genuine robustness property. (ii) Safety is no longer guaranteed — (18) is not an exact penalty, so a finite W_{obs} admits a finite violation, and Section 3.3.3 makes that statement quantitative. A control-barrier formulation [1] would recover the guarantee, at the price of feasibility and of a harder NLP. (iii) The penalty is evaluated only at the $N + 1$ discrete nodes, so nothing forbids the continuous inter-sample trajectory from clipping a thin obstacle.

Point (iii) is usually left as a caveat. It is worth checking instead, because it is a one-line computation. Between two consecutive nodes the robot advances at most $v_{x,\text{max}} \Delta t$, against a keep-out radius r_{obs} , so the penalty is meaningful only while

$$\frac{v_{x,\text{max}} \Delta t}{r_{\text{obs}}} < 1. \quad (19)$$

On the deployed profile the numerator is 0.40×0.35 , i.e. about 14 cm against 0.40 m of radius, so the ratio is roughly 0.35: below one, but not by the margin the phrase “comfortably” would suggest, and it has grown by half since the envelope was widened. The check is reported here rather than assumed because it is not invariant. It couples two parameters that are tuned for unrelated reasons — the sampling time of Section 5.5 and the input envelope of Table 1 — and lengthening Δt to buy horizon, which is the cheapest way to buy it, walks the ratio towards one. Above one the obstacle term is not merely weak, it is absent between nodes, and the geometric avoidance falls entirely to the A^* layer on the inflated grid without anything in the report saying so.

Sentinel padding. M is fixed. When fewer than M returns lie inside the search radius, the missing rows of the obstacle parameter are filled with a sentinel at 10^3 m, whose contribution to (18) is numerically zero. This apparently cosmetic detail is what allows the NLP to be built *once*: the number of terms, and therefore the sparsity pattern of the Jacobian and Hessian, never changes between cycles, so the symbolic factorisation and the warm start remain valid regardless of how cluttered the scene is.

3.3.3 The obstacle as a true constraint: exact ℓ^1 penalty

The alternative to a penalty in the cost is a constraint with a slack. The deployed code implements both, selectable by a parameter: the obstacle becomes

$$d(x_k, p_j) \geq d_{\text{safe}} - s_{jk}, \quad s_{jk} \geq 0, \quad k = 1, \dots, N, \quad (20)$$

with the slack penalised in the cost either linearly ($\rho_s \sum s$) or quadratically ($\rho_s \sum s^2$). The question this settles is the one Section 3.3.2 could only raise: at what weight, if any, does the relaxation stop being a relaxation.

The exact-penalty theorem answers it. For a non-smooth ℓ^1 penalty there exists a finite threshold — the largest multiplier of the corresponding hard constraint — above which the residual violation is exactly zero, whereas an ℓ^2 penalty decays like $1/\rho_s$ and never vanishes at any finite weight. Both predictions are testable here, because the multiplier is available from the problem itself.

Solved as a hard constraint on a narrow-gap scenario with $d_{\text{safe}} = 1.10$ m — chosen so that the constraint genuinely bites — the instance is feasible and the largest multiplier of an active distance constraint is $\mu^* = 26767$.

Table 5: Residual constraint violation against the penalty weight, for the non-smooth and the smooth relaxation of the same distance constraint. Bold: the violations that are exactly zero, which is the result the exact-penalty theorem predicts and the smooth relaxation never attains. Zero entries are below the solver tolerance of 1×10^{-8} m.

ρ_s	max slack, ℓ^1 [m]	max slack, ℓ^2 [m]
1×10^3	5.60×10^{-1}	6.53×10^{-1}
1×10^4	1.44×10^{-1}	3.19×10^{-1}
1×10^5	0	9.54×10^{-2}
1×10^6	0	1.33×10^{-2}
1×10^7	0	1.34×10^{-3}
1×10^8	0	1.34×10^{-4}

The measurement matches the theorem on both branches. The ℓ^1 slack is a threshold phenomenon: nonzero below, and zero to solver tolerance from $\rho_s = 1 \times 10^5$ onwards — which is the first sampled weight above μ^* . The ℓ^2 slack decays with a fitted log-log slope of -1.00 against a predicted -1 , and is still nonzero five decades later. The practical consequence is that ρ_s stops being a hyperparameter: it is read off the multipliers of the problem rather than tuned.

Two defects of formulation found by implementing it. Both have theoretical content and both are worth reporting, because each is a way to make a well-posed problem unsolvable.

First, **the constraint must not be imposed at $k = 0$** , for the reason already given in Section 2.6: there the state is fixed by an equality, the constraint depends on no decision variable, and imposing it makes the program infeasible exactly when the robot is already inside d_{safe} — that is, exactly when one would want it. Measured on a recorded cycle, the slack saturated at precisely the violation present at $k = 0$ and the solver could do nothing about it.

Second, **the residual infeasibility on real cycles is a property of $\mathcal{U}$, not of the data**. Even starting from $k = 1$, a demanding d_{safe} was found infeasible on recorded cycles: the admissible set is anisotropic, the robot advances largely along its own heading, and it cannot back away laterally from a wall of LiDAR points as fast as it approached it. This is the interaction between the asymmetry of the admissible input set and a state constraint, and it reappears in Sections 4.5 and 3.3.7 from two other directions.

This particular result is due a re-measurement. The observation above was made when the lateral bound was an order of magnitude below the forward one. The deployed envelope has since been widened to $v_{y,\text{max}} = 0.20$ m/s against $v_{x,\text{max}} = 0.40$ m/s — a ratio of one to two, not one to fifteen — and at that ratio the robot has real lateral authority. The mechanism as stated is therefore weaker than it was, and how much of the infeasibility survives is an open question rather than a reported result: the number is not regenerated by the campaign of Section 5.1, and it is flagged here rather than quietly carried forward. The threshold result of Table 5, which is the substance of this section, is unaffected: it is measured on the synthetic instance at $d_{\text{safe}} = 1.10$ m under the current profile, and there the instance is feasible and the ℓ^1 slack does reach zero.

What it would cost to deploy. Switching the obstacle from the cost to the constraints takes the inequalities from 60 to 300 and the iteration count up by a factor of several. The deployed iteration cap would

not suffice. This is also what makes the solver comparison of Section 4.4 possible at all: the same system can be placed in two regimes that differ by a factor five in the number of inequalities, which is exactly the axis along which the standard advice on solver choice is stated.

3.3.4 Constraints

The deployed program carries only equality constraints and input bounds:

$$x_{k+1} - f(x_k, u_k) = 0, \quad k = 0, \dots, N-1, \quad (21a)$$

$$x_0 - \hat{x} = 0, \quad (21b)$$

$$v_{x,\min} \leq u_{1,k} \leq v_{x,\max}, \quad k = 0, \dots, N-1, \quad (21c)$$

$$|u_{2,k}| \leq v_{y,\max}, \quad |u_{3,k}| \leq \omega_{\max}, \quad k = 0, \dots, N-1. \quad (21d)$$

Three remarks.

No state constraints. Neither the obstacle clearance nor the velocity states are constrained. The former is penalised. The latter are bounded implicitly, but the argument needs stating with its hypothesis attached, because it is not unconditional: (13) is a convex combination of a bounded command and the previous velocity, so $v_{x,k}$ stays in $[\min(v_{x,0}, v_{x,\min}), \max(v_{x,0}, v_{x,\max})]$ — the input box only if $v_{x,0}$ is already inside it. It need not be: $x_0 = \hat{x}$ is an estimate, obtained from filtered pose differences, and nothing constrains an estimate to the set of admissible commands. On the deployed profile the point is moot for $k \geq 1$, because the degenerate lag of Section 3.2 collapses (13) to $v_{k+1} = u_k$ and the prediction lands in the box after a single step whatever the estimate says; on hardware, with a real τ , the first few nodes would inherit the excess and decay towards the box rather than sit in it. The feasible set is in any case a box intersected with an affine-in- X manifold, and a feasible point is always trivially available.

The input box is asymmetric, and deliberately so. Constraint (21c) admits reversing, but not at the speed of forward travel: $v_{x,\min}$ is a fraction of $v_{x,\max}$, and the lateral bound is smaller still. None of the three asymmetries is a safety derating applied on top of a symmetric machine, and it is worth separating what causes each. The lateral bound is *kinematic*: a biped strafes genuinely more slowly than it walks forward. The backward bound is *behavioural*: backing away is a disengagement manoeuvre, and keeping it slower than forward travel is what stops the optimizer from adopting it as a cheap shortcut at every stop. And on hardware the backward bound would be $v_{x,\min} = 0$, a hard non-negativity, because the LiDAR leaves a blind cone behind the robot and reversing means moving blind — a constraint that the simulated sensor of Section 5.1, which samples the full azimuth, does not impose. The deployed value is therefore a property of the test bench and not of the platform, and it should be read as such. It is also, as Sections 3.3.3 and 4.5 show, the part of the formulation that most restricts what any state requirement can ask of the robot.

Bounds are parameters, not constants. The three velocity limits enter the NLP as CasADi parameters, so a supervisor can shrink them at runtime without rebuilding the problem.

3.3.5 Terminal ingredients and recursive feasibility

The deployed formulation carries no terminal set and no terminal constraint, so none of the standard guarantees of nominal stability and recursive feasibility [3, 5] applies. Recursive feasibility is nevertheless *structurally* present, for the reason given in Section 3.3.4: with no state constraints, every input sequence in the box is admissible, so the NLP always has a solution and the closed loop can never be aborted by infeasibility. What is lost is the guarantee that the closed loop converges, and it is worth naming the shape the loss takes: the tracking term can decrease along a trajectory that walks into a well of the obstacle penalty, and nothing but that penalty stops it. Feasibility that costs nothing to keep is not the same as a descent property.

That is the situation to improve on, and the deployed code implements the improvement: a terminal equilibrium constraint $v(N) = 0$ — there must exist a braking trajectory inside the horizon — relaxed by a slack and penalised in ℓ^1 norm, so that by the result of Section 3.3.3 the constraint is effectively hard whenever it is satisfiable and yields gracefully when it is not.

Across the cycles replayed from a recorded mission the terminal slack never leaves zero (maximum 0, always admissible: yes), and the cost of imposing it, measured as the relative increase of the optimal objective, ranges from 1.2% to 40.4%. The upper end is paid exactly where the objective was small, that is where the robot was travelling well and now has to predict its own braking — which is the expected trade: a terminal constraint buys the stability argument by spending part of the horizon on stopping.

Why this result must be reported with its cause attached. That the slack is always zero is not, by itself, evidence that the constraint works: an unimplemented constraint would report the same thing. The reason it is zero here is the degeneracy of Section 2.5. With $\nu = 1.000000$ the model reduces to $v_{k+1} = u_k$ and

the robot reaches zero velocity in a *single step*, so the terminal set is trivially reachable from anywhere and the constraint costs nothing in admissibility. A counter-test with realistic time constants confirms the mechanism: as τ grows the slack becomes strictly positive and grows with it, because the horizon is no longer long enough to stop. On hardware, with an identified τ , this constraint is the one that would decide the maximum speed from which the robot can still be brought to a halt inside the horizon — which is the physical reading of the terminal feasible set, and the form in which it would actually bind.

3.3.6 Reference trajectory: time-parametrised against path-parametrised

The deployed reference advances along the smoothed A^* path at a constant cruise speed,

$$s_k = \min(s_0 + v_{\text{ref}} k \Delta t, L_{\text{path}}), \quad (22)$$

interpolating the position inside the containing segment and taking the heading as the segment tangent.

The clip against L_{path} in (22) is not a guard against indexing past the end of an array. When the remaining path is shorter than $v_{\text{ref}} N \Delta t$ the reference saturates at its last point, every remaining node asks for the same position, and the tracking term turns into a terminal-position term: that is the mechanism by which the controller decelerates into the local goal, and the formulation has no other. On the deployed profile the horizon covers about a metre and a half against a window six metres across, so the saturation is inactive in transit and engages only on approach — which is also the only moment at which the terminal equilibrium constraint of Section 3.3.5 has something to ask for.

The cruise speed is set below $v_{x,\text{max}}$ deliberately: with $v_{\text{ref}} = v_{x,\text{max}}$ the tracker saturates permanently and never settles onto the path. The price is structural: 32% of the available forward speed is unreachable *by construction*, and no cost weight can recover it, because the reference never asks for it.

The alternative the theory offers is to make the path abscissa a decision variable: the tracker chooses how far to advance rather than being told. The deployed code implements this too, with $\theta(0) = 0$, $\Delta\theta \geq 0$, $\theta(N) \leq 1$ and a progress term $\alpha_3(1 - \theta)^2$ in the cost. The reference $\bar{z}(\theta)$ is represented by a polynomial whose coefficients are parameters, refitted at every solve on the A^* path — necessary because θ is now a decision variable and IPOPT is a Newton-type method, so a piecewise-linear reference through waypoints would have a discontinuous second derivative at every node.

Table 6: Time-parametrised against path-parametrised reference, averaged over 6 cycles replayed from the recorded run `industrial_v6`.

quantity	time-parametrised	path-parametrised
mean v_x [m/s]	0.269	0.379
advance over the horizon [m]	1.351	2.038
IPOPT iterations	9.3	25.2

Over 6 moving cycles replayed from the recorded run, the mean commanded forward speed rises from 0.269 to 0.379 m/s and the distance covered within one horizon from 1.351 to 2.038 m: the robot advances 51% further at the same horizon, and the solver saturates $v_{x,\text{max}}$ on nearly every step. The cruise-speed parameter does not reappear in another guise — θ is a variable, not a hyperparameter.

The cost is $9.3 \rightarrow 25.2$ IPOPT iterations, an increase of about 70%: the program has $N + 1$ more variables and N more inequalities, and it is less well conditioned. Reported as a trade rather than as an improvement, this is the cleanest reformulation available to the stack.

A trap worth documenting. On cycles where the robot is stationary the comparison is meaningless: θ advances without the robot being able to follow, and with a strong progress weight one observes $\theta \rightarrow 1$ while the robot stands still — the reference detaches from the dynamics. The comparison is therefore restricted to cycles with genuine motion. This is not a numerical artefact but a modelling statement: a path-parametrised reference is only a reference if the dynamics can realise the progress it encodes.

Warm start. Whichever parametrisation of the reference is used, the previous solution is reused as the initial guess by the standard shift-and-append rule

$$U_{\text{new}}^{(0)} = [u_1^*, \dots, u_{N-1}^*, u_{N-1}^*], \quad X_{\text{new}}^{(0)} = [x_1^*, \dots, x_N^*, x_N^*], \quad (23)$$

which is consistent to first order because the reference itself shifts by one step. Two safeguards protect it: the guess is discarded after three consecutive solver failures, and it is discarded whenever the optimal cost exceeds

five times the mean of the last seven solves. The second rule is the interesting one, and it is a statement about the geometry of the problem rather than about numerics: a cost spike signals that the reference has changed homotopy class — a new wall has entered the window, and the A^* path now passes on the other side of it — in which case the shifted guess lies in the basin of the *old* solution and would either slow the solver down or drag it to the wrong minimum. Section 5.4 shows, in the bifurcation sweep, the situation the second safeguard exists to catch; its computational benefit is not re-measured in this report.

3.3.7 Robust constraints: tightening from the measured prediction error

Every constraint in (9) is imposed on the *predicted* trajectory, and Section 5.3 measures how far the prediction drifts from what the robot actually does. A clearance guaranteed in the plan is therefore not a clearance guaranteed on the floor. The standard remedy is constraint tightening: replace d_{safe} with a margin $\gamma(k)$ that grows along the horizon,

$$\|p_k - o_j\| \geq d_{\text{safe}} + \gamma(k) - s_{jk}. \quad (24)$$

The particularity of this project is that $\gamma(k)$ need not be postulated. It is read off the 95th percentile of the prediction error recorded in the bags, so the tube is derived from data rather than from an assumption about the disturbance.

Table 7: Constraint back-off derived from the 95th percentile of the measured prediction error. The offset at $k = 0$ is subtracted: it is time misalignment, not model uncertainty, and including it would inflate the tube by a constant.

k	$k \Delta t$ [s]	$\gamma(k)$ [m]
0	0.0	0.0000
1	0.3	0.0820
2	0.7	0.1218
3	1.0	0.2018
4	1.4	0.2879
5	1.8	0.3597
6	2.1	0.4305
7	2.4	0.5075
8	2.8	0.5657
9	3.1	0.6521
10	3.5	0.7242
11	3.8	0.7956
12	4.2	0.8537
13	4.5	0.9237
14	4.9	0.9915
15	5.2	1.0363

Three properties hold by construction and were checked rather than assumed. $\gamma(0) = 0.0000$ exactly, because at the first node the state is fixed by an equality and there is nothing to hedge against — the constraint is not tightened where tightening could only remove admissible motion. The margin is monotone (yes), as uncertainty is. And it reaches 1.0363 m at the end of the horizon, the same order as the safety radius itself. The offset at $k = 0$ is subtracted before fitting, since Section 5.3 shows it to be time misalignment rather than model error, and including it would inflate the whole tube by a constant.

Table 8: Effect of the tightening on the predicted trajectory.

scenario	d_{safe}	clear. without	clear. with	Δ	slack w/o / with	outcome
narrow_gap	0.40	0.4509	0.3241	-0.1268	0 / 0.690	infeasible
narrow_gap	0.70	0.7000	0.9989	0.2989	0 / 0	tightening effective
narrow_gap	1.00	1.0000	1.1462	0.1462	0 / 0	tightening effective
u_trap	0.40	0.9825	1.2555	0.2730	0 / 0	tightening effective
u_trap	0.70	0.9825	1.2963	0.3138	0 / 0	tightening effective
u_trap	1.00	0.9825	1.4191	0.4365	0 / 0	tightening effective

Measured on the predicted trajectory, the outcomes separate cleanly. Where $d_{\text{safe}} + \gamma$ stays below the clearance the trajectory already keeps, the constraint is inactive and correctly does nothing. Where it bites, the predicted

clearance increases by up to 0.437 m with the slack still exactly zero: the margin is *respected*, not violated and paid for — which is the whole point of pairing the tube with the ℓ^1 relaxation of Section 3.3.3, since a tube that asks more than $\mathcal{U}$ can deliver makes the penalty yield instead of making the program infeasible.

The limit of this measurement, and what it says about the architecture. The effect is *not* observable in the closed loop of this harness: the executed clearance comes out identical with and without tightening. The reason is the architecture described in Section 2.1 — the loop is closed by tracking a look-ahead setpoint on the predicted trajectory with a proportional law, which washes out fine differences between MPC solutions. The tightening guarantees the margin *in the plan*, and that is where it has been verified. This is not a limitation of the technique but of the test bench, and it is the strongest argument in the report for applying u_0^* directly, as the receding-horizon principle prescribes.

On the recorded industrial run the tightening is inert for a different and more instructive reason: at the tightest cycles the robot sits about 0.6 m from the obstacles and cannot move away. Raising d_{safe} grows the slack but does not move the trajectory. The binding constraint is not the threshold, it is $\mathcal{U}$ — the same asymmetry found in Section 3.3.3 and quantified in Section 4.5. Three independent measurements arrive at the same conclusion, which is the strongest form the report has for saying that the input set, not the cost, is where this system is limited.

3.3.8 On the absence of integral action

The formulation has no integral action and no disturbance estimator: the state (6) is not augmented with an input-disturbance model, and the reference is tracked in a purely proportional–predictive sense. For a nominal plant this is immaterial, but any constant unmodelled input — a mis-identified τ , a lateral drift of the gait, a slope — produces a persistent offset from the path rather than being integrated away. The offset is therefore a property of the formulation and not of the tuning, and no choice of Q removes it. Section 5.3 measures it; Section 6 discusses the standard remedy, state augmentation with an integrated tracking error as in the velocity-form MPC of [7].

4. Optimization procedure

4.1. Problem dimensions and structure

Table 9 reports the size of the NLP against the horizon, measured on the deployed profile rather than counted by hand.

Table 9: Size and sparsity of the NLP against the prediction horizon, from the deployed profile. Bold: the deployed configuration.

N	vars	eq.	bounds	params	jac nnz	jac dens. [%]	hess nnz	hess dens. [%]
10	96	66	40	91	256	2.52	600	6.51
15	141	96	60	121	381	1.73	885	4.45
25	231	156	100	181	631	1.07	1455	2.73
50	456	306	200	331	1256	0.54	2880	1.39

At the deployed $N = 15$ the program carries 141 decision variables and 156 constraints, of which 96 are equalities — the dynamics plus the initial condition — and 60 are simple bounds on the inputs. A further 121 quantities enter as CasADi parameters: the measured state, the reference, the sentinel-padded obstacle positions and the three velocity bounds. The problem is therefore medium-scale, sparse, smooth and non-convex.

The constraint Jacobian is 1.73% dense and the Hessian of the Lagrangian 4.45%. Both densities fall as N grows while the nonzero counts grow linearly, which is the signature of the simultaneous parametrisation: the Jacobian of (21a) is block bi-diagonal, the Hessian block tri-diagonal with the off-diagonal input blocks coming from the rate penalty, and nothing couples distant stages. The linear-algebra cost of one interior-point iteration is consequently linear in N .

It is worth being explicit about what is *not* in this table, because it shapes everything in Section 4.4. There are no obstacle constraints: the $2M(N + 1)$ obstacle terms all live in the objective. There is therefore no obstacle multiplier, no non-trivial active set to identify, and the entire inequality budget of the deployed program is 60 box bounds on the inputs.

4.2. Derivative computation

Every quantity the solver needs — the gradient of the objective, the Jacobian of the constraints, the Hessian of the Lagrangian — is generated symbolically by CasADi and evaluated by generated code. The alternative is finite differences, and the comparison is not a formality: it decides whether the second-order method of Section 4.3 is affordable at all.

Table 10: Accuracy of the finite-difference gradient against the step size, measured against the AD gradient. The optimum sits at $h = 1.49 \times 10^{-8}$ one-sided (theory: $\sqrt{\varepsilon} = 1.49 \times 10^{-8}$) and $h = 6.06 \times 10^{-6}$ central (theory: $\varepsilon^{1/3} = 6.06 \times 10^{-6}$).

step h	rel. error, one-sided	rel. error, central
1.00×10^{-4}	9.18×10^{-4}	2.15×10^{-8}
6.06×10^{-6}	5.56×10^{-5}	9.84×10^{-11}
1.00×10^{-6}	9.17×10^{-6}	5.29×10^{-10}
1.49×10^{-8}	1.27×10^{-7}	3.77×10^{-8}
1.00×10^{-10}	9.66×10^{-6}	4.36×10^{-6}

At a representative solve the objective has 141 variables. One evaluation costs 149.0 μ s and one full gradient by reverse-mode algorithmic differentiation (AD) 398.6 μ s, i.e. 2.41 objective evaluations — a small constant, and crucially one that does not grow with the number of variables. The same gradient by finite differences costs 142 evaluations one-sided and 282 central, both growing linearly with n , and is less accurate at every step size: the best relative error attainable is 1.27×10^{-7} one-sided and 9.84×10^{-11} central, against machine precision for AD. The measured optimal steps coincide with the theoretical $\sqrt{\varepsilon_M}$ and $\varepsilon_M^{1/3}$, ε_M being the machine epsilon. For a real-time loop the cost settles the question: central differences alone would consume 34% of the 125 ms cycle budget, for a gradient that is worse, and the fraction grows linearly with the horizon. The ratio itself is a micro-benchmark on ~ 100 μ s timings and its second digit is not meaningful; what is stable, and is the point, is that it remains a small constant.

The exact Hessian, and what it buys. Because AD supplies exact second derivatives at a cost comparable to the first, the full Newton system can be used rather than a quasi-Newton approximation.

Table 11: Interior-point iterations with the exact Hessian and with the limited-memory quasi-Newton approximation, on the same instance. Bold: the lower iteration count.

Hessian	iterations	solve [ms]	J^*	status
exact Hessian	35	677.7	3646	Solve_Succeeded
L-BFGS	89	1763.2	3646	Solve_Succeeded

On the same instance the exact Hessian converges in 35 iterations against 89 for the limited-memory quasi-Newton approximation — a 61% reduction — and both reach the same objective value. The total time is closer than the iteration counts suggest, because each quasi-Newton iteration is cheaper: this is the Newton / quasi-Newton trade-off in its textbook form, measured without writing a solver. Section 4.4 shows the other side of the same coin, where the exact Hessian of a non-convex program produces an indefinite quadratic program (QP) subproblem.

4.3. Solver methodology

The NLP is solved with IPOPT [11], a primal–dual interior-point method with a filter line search. Writing the problem as $\min_z \phi(z)$ subject to $c(z) = 0$ and $z^L \leq z \leq z^U$, the bounds are replaced by a logarithmic barrier

$$\varphi_{\mu_b}(z) = \phi(z) - \mu_b \sum_i \left[\ln(z_i - z_i^L) + \ln(z_i^U - z_i) \right], \quad (25)$$

the resulting equality-constrained problem is solved approximately by a Newton step on the perturbed Karush–Kuhn–Tucker (KKT) system

$$\begin{bmatrix} \nabla_{zz}^2 \mathcal{L} + \Sigma & \nabla c(z)^\top \\ \nabla c(z) & 0 \end{bmatrix} \begin{bmatrix} \Delta z \\ \Delta \lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_b} \\ c(z) \end{bmatrix}, \quad (26)$$

and μ_b is driven to zero. Each iteration requires one factorisation of the sparse symmetric indefinite matrix in (26), performed by MUMPS on the pattern of Section 4.1.

There is a pleasing coincidence between (25) and (18) that is worth naming: the obstacle term of the deployed program is itself a barrier, and α_{obs} plays the role of an inverse barrier parameter. Steeper means better avoidance and worse conditioning, which is exactly the trade-off μ_b governs inside the solver. The difference is that IPOPT drives its own barrier parameter to zero along a central path, whereas α_{obs} is fixed by the tuning — so the program is solved with a barrier that never relaxes.

4.4. Interior point against active set

The standard advice is that active-set methods win when the inequalities are few and interior-point methods win when they are many. This stack can be moved between the two regimes without changing anything else, because the obstacle formulation is switchable: with the obstacles in the cost the program carries 60 inequalities, and with the obstacles as genuine constraints (Section 3.3.3) it carries 300. That makes the advice testable rather than quotable.

Table 12: The same instance solved by a primal–dual interior-point method and by an SQP with an active-set QP solver, in both obstacle regimes. Both are started cold. Bold: the faster of the two.

regime	ineq.	IPOPT ms / iter	SQP ms / iter	speed-up	same min.
penalty (obstacles in the cost)	60	67 / 20	146 / 6	2.2×	yes
penalty (obstacles in the cost)	60	89 / 23	860 / -1	9.6×	no
penalty (obstacles in the cost)	60	225 / 77	275 / 7	1.2×	yes
ℓ^1 (obstacles as constraints)	300	146 / 35	9307 / 16	63.6×	no
ℓ^1 (obstacles as constraints)	300	173 / 46	13204 / -1	76.4×	no
ℓ^1 (obstacles as constraints)	300	162 / 45	11062 / -1	68.2×	no

The rule does not reproduce: the active-set solver does not win even in the small regime, where the interior-point method is already 2.2×

put. Two cautions, without which the numbers would be misleading. First, the real advantage of an active set in MPC is warm starting *between consecutive solves*: the active set changes by a few rows per cycle and the factorisation is reused, which is the online active-set strategy the method was designed for. Both solvers are started cold here, deliberately, so as to favour neither — which removes from the active-set method precisely what makes it competitive in a receding-horizon loop. Second, the advice is stated for convex programs, and this one is not.

A by-product that the theory predicts. With the exact Hessian of the Lagrangian, the QP subproblem of the sequential quadratic programming (SQP) method is repeatedly flagged indefinite. This is expected and instructive: a non-convex program can produce a non-convex QP, and a non-convex QP has no unique solution. It is exactly the reason a Gauss–Newton Hessian — positive semi-definite by construction — is the standard recommendation for SQP, and it is the counterpart of the result in Section 4.2, where the same exact Hessian was the better choice inside an interior-point method. The two are not in contradiction: an interior-point method regularises the indefinite system in (26), an SQP must solve the QP.

A methodological note. On a first run the SQP appeared to converge in a single iteration to a worse objective than IPOPT. Comparing the time of two solves that terminate at *different* minima means nothing, and the tool now verifies that the two objectives agree before declaring a winner. Table 12 reports that check in its last column.

4.5. Optimality conditions on the deployed problem

Everything above assumes that what the solver returns is a solution. That is a verifiable statement, and this section verifies it on cycles extracted from a recorded mission rather than on a synthetic instance.

At 9 sampled cycles, the linear independence constraint qualification (LICQ) holds at every one (yes) — the rank of the Jacobian of the active constraints equals their number — so the KKT multipliers exist and are *unique*. Strict complementarity holds at every one (yes), which means the critical cone coincides with the null

Table 13: Optimality diagnostics at sampled cycles of the recorded run `industrial_v6`. “active” counts equalities and active bounds together; “rank” is the rank of the Jacobian of the active constraints, so LICQ holds when the two coincide.

cycle	t [s]	active	of which bounds	rank	LICQ	cone dim.	$\lambda_{\min}$ proj.	
0	0	103		7	103	yes	38	1.33×10^0
86	46	101		5	101	yes	40	1.26×10^0
173	85	98		2	98	yes	43	1.25×10^0
260	120	101		5	101	yes	40	1.31×10^0
347	153	97		1	97	yes	44	1.24×10^0
433	191	100		4	100	yes	41	1.26×10^0
520	245	100		4	100	yes	41	1.25×10^0
607	315	101		5	101	yes	40	1.28×10^0
694	358	98		2	98	yes	43	1.23×10^0

space of the active Jacobian and the second-order condition can be checked exactly rather than conservatively. And the reduced Hessian is positive definite on that cone at every cycle (yes), the smallest projected eigenvalue over the profile being 1.23×10^0 . Taken together these are a certificate of strict local optimality, which is the most that can be claimed for a non-convex program.

The result worth reporting is the shape of the active set. The critical cone holds between 38 and 44 dimensions across the sampled cycles — 38 at the first, 43 at the last — and does not collapse: with 141 variables and 98 active constraints at the end, the freedom the controller has when it arrives is the freedom it had when it started. Almost every active row is an equality of the dynamics; the input bounds contribute a handful per cycle and never more. The controller is cost-driven for the whole mission.

This reverses what earlier campaigns of this report measured, and the cause is the input envelope rather than the analysis. On the previous profile — lateral speed an order of magnitude below forward travel, reversing forbidden — the cone fell from 45 dimensions to 8, the lateral and yaw limits saturated on essentially every step of the horizon, and the non-negativity of v_x on nearly all of them, so that the controller ended the mission undergoing the trajectory rather than choosing it.² Widening the envelope to $|v_y| \leq 0.20$ m/s and $v_x \in [-0.20, 0.40]$ m/s was the recommendation that followed from them, and the table above is the run that acted on it. The input set is no longer what decides the motion, and the claim this section used to carry — that no choice of cost weights could return what the input set had taken away — no longer holds on the deployed profile.

Two traps in reading multipliers out of a modelling layer. Both were met here and both would have produced quietly wrong numbers. CasADi’s `Opti` canonicalises every constraint as $\text{lbg} \leq g(z) \leq \text{ubg}$ and *absorbs into the bounds* any parametric right-hand side, so $X_{:,0} = \hat{x}$ becomes $g = X_{:,0}$ with $\text{lbg} = \text{ubg} = \hat{x}$: evaluating $|g|$ on those rows returns the state, not the residual, which is always $g - \text{lbg}$. And two distinct rows can share the same expression with different bounds — $u_{1,k} \geq 0$ and $u_{1,k} \leq v_{x,\max}$ are both the row $u_{1,k}$ — so activity must be decided on the distance from the bound, not on $|g| \approx 0$, or both sides of every box are counted active.

4.6. Implementation and configuration

The program is built once, symbolically, and every quantity that changes between cycles is a parameter rather than a constant: the measured state, the reference, the sentinel-padded obstacle positions and the three velocity bounds. Consequently the expression graph, the derivative code and the sparsity patterns are generated exactly once at start-up, and a cycle only writes numbers into parameter slots and calls the solver. The graph is flattened so that the interpreter overhead of a problem with thousands of elementary nonlinear operations does not dominate. The first solve after start-up therefore pays a one-off construction cost, which is the single deadline violation of an entire run and is absorbed by holding the robot still during initialisation.

The solver options are deliberately austere, with a low iteration cap: a non-converged iterate is still usable as a reference by the downstream setpoint stage, which is a property of the architecture of Section 2.1 and would not hold if u_0^* were applied directly. The real-time protection lives in the supervisor rather than inside IPOPT — a solver whose behaviour depends on a wall-clock limit is not deterministic and cannot be characterised. The

²Those two figures are historical: they were measured on the narrow envelope and are quoted from that run, not recomputed. Every other number in this section comes from the macros of the current campaign.

supervisor clears the warm-start cache on solver failure, falls back to a zero command after repeated failures, and can shrink the velocity bounds through the parameter interface when the failure rate rises.

5. Performance analysis

5.1. Experimental setup

All results in this section are produced by an offline harness that imports the *production* modules — the grid, the A^* planner and the MPC tracker — so every number is generated by the deployed formulation and the deployed solver, not by a re-implementation. Two kinds of experiment are used and they answer different questions.

Replay from recorded missions. The bag `industrial_v6` contains a full run of the G1 in the industrial world: the poses, the LiDAR scans, the A^* paths and the MPC predictions as they were published. Replaying it gives real operating points, with the geometry and the clutter the robot actually met, and is what Sections 5.3 and 4.5 use. It cannot answer counterfactual questions.

Closed-loop synthetic scenarios. For counterfactuals — what if the horizon were shorter, what if the weights were different — the harness runs full missions in scripted worlds, with a ray-marched planar LiDAR and the discrete model of Section 3.2 as the plant. This is what Sections 5.5, 5.6 and 5.7 use. Two of these worlds recur: a narrow gap and a U-shaped trap.

A third family is used only in Section 5.9, because it probes a different layer. The worlds `long_wall`, `horseshoe` and `dead_end` each place a *non-convex* obstacle between the robot and the goal — a wall long enough that its ends leave the window, a horseshoe, and a corridor closed at the far end. They are run with *limited perception*: the robot sees only out to the LiDAR range and only what is not occluded, and accumulates what it sees. That last detail is not a refinement but the whole experiment, because with the obstacle known in advance the discrete planner never enters the trap and the failure under study does not occur.

A correction to an earlier reading of the second family, which the report would be weaker for omitting. Previous versions of this section reported that in the U-trap the minimum clearance came out at zero in essentially every configuration tested, and concluded that the world discriminated on time and computation but said nothing about safety. That reading was an artefact of the harness, not a property of the world: the replay loop selected its setpoint at a look-ahead distance that no longer matched the deployed one, so the fallback proportional law took over and the output of the MPC was discarded on every cycle. With the look-ahead read from the profile, the grazing configurations are confined to the very short horizons — 7 of the 25 combinations in the U-trap, all of them at $N\Delta t \leq 2$ s, and none at all in the narrow gap — and the median clearance is 0.856 m. The clearance axis therefore does discriminate, and Section 5.5 reads it. The general lesson is worth keeping: a harness that silently bypasses the controller under test reports plausible numbers, and the only thing that exposed it was a quantity — clearance — that had no business being identical across configurations.

5.2. The cost landscape, and why this is not a potential field

Before the campaigns, it is worth looking at the object all of them are about. Fig. 6 draws, over the world plane of the recorded industrial run, the cost of *being* at a point,

$$c(p) = \underbrace{\text{tracking cost of the reference}}_{\text{the } Q \text{ block of Eq. (17)}} + \underbrace{J_{\text{obs}}(p)}_{\text{Eq. (18)}}, \quad (27)$$

with the executed trajectory, the A^* reference and one MPC horizon drawn on the surface, and the per-cycle optimal cost J^* underneath.

The distinction that figure makes visible is the one between c and J , and it is the reason the stack uses an optimizer rather than a reactive law. The quantity plotted is c , the cost of occupying a point; what the program minimises is

$$J(U) = \sum_{k=0}^N c(p_k) + \text{input terms}, \quad (28)$$

the cost of a whole *trajectory*. A controller that descended c pointwise would be an artificial potential field, and it would stop at the first local minimum of the surface. The MPC minimises the sum along a horizon instead, so it can — and in panel (a) visibly does — move *uphill* on c when that buys a lower total. This is what lets the same cost function carry the robot out of a concave pocket, and it is the sense in which the obstacle term of Eq. (18) is a penalty inside an optimization problem rather than a repulsive force.

One further reading, which anticipates Section 4.5. The shaded region in panel (b) is not a level set of the cost: it is the set of points the robot can reach within one horizon given $\mathcal{U}$. Most of the plane is excluded from the

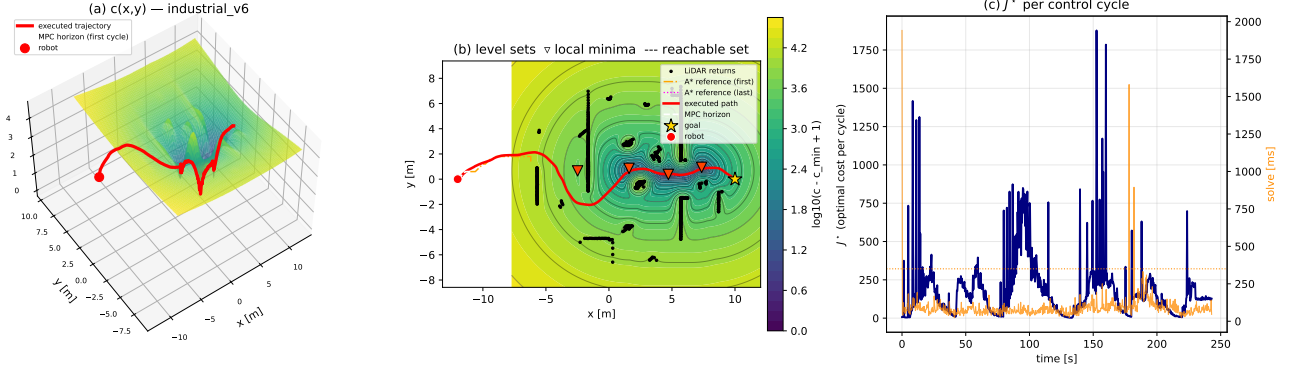


Figure 6: The navigation cost landscape of the recorded run `industrial_v6`. (a) the surface $c(p)$ of Eq. (27) over the world plane, with the executed trajectory (red) and one MPC horizon (dashed) drawn *on* the surface: the obstacle term raises a ridge along every wall of the plant, the tracking term forms the valley along the A^* reference. (b) the same field as level sets, with the local minima marked and the shaded region showing the set reachable within one horizon under $\mathcal{U}$ — the constraint, not the cost, is what excludes most of the plane. (c) the optimal cost J^* of each control cycle, and the solve time beside it. The point worth noticing is in panel (a): *the trajectory does not follow steepest descent*, and climbs the surface wherever doing so lowers the sum over the horizon.

decision by the *constraints*, not by the objective — and on this platform, with $v_{y,\max} = 0.20$ m/s against a forward bound of 0.40 m/s and reversing admitted only at a fraction of that, the region is markedly asymmetric about the heading. The cost is what chooses inside it; the input set is what draws it.

5.3. Prediction error: the model against the plant

The most consequential measurement in this report requires no new experiment: the bags already contain, at each cycle, the trajectory the MPC predicted, and the poses the robot subsequently reached. Comparing the prediction published at time t with the pose measured at $t + k\Delta t$, over 695 usable cycles, gives the growth of the prediction error along the horizon.

Table 14: Open-loop prediction error along the horizon, over 695 cycles of the recorded run `industrial_v6`.

k	$k \Delta t$ [s]	median error [m]	95th pct. [m]
0	0.0	0.0561	0.0758
1	0.3	0.0718	0.1578
2	0.7	0.1185	0.1976
3	1.0	0.1690	0.2776
4	1.4	0.2098	0.3637
5	1.8	0.2665	0.4356
6	2.1	0.3297	0.5063
7	2.4	0.3949	0.5833
8	2.8	0.4572	0.6416
9	3.1	0.5175	0.7279
10	3.5	0.5792	0.8000
11	3.8	0.6353	0.8714
12	4.2	0.6937	0.9295
13	4.5	0.7548	0.9995
14	4.9	0.8148	1.0673
15	5.2	0.8531	1.1121

The residual at $k = 0$ is 0.0561 m and is *not* model error: at the first node the predicted state *is* $\hat{x}$, imposed as an equality constraint. It measures the delay between the instant the prediction is published and the instant

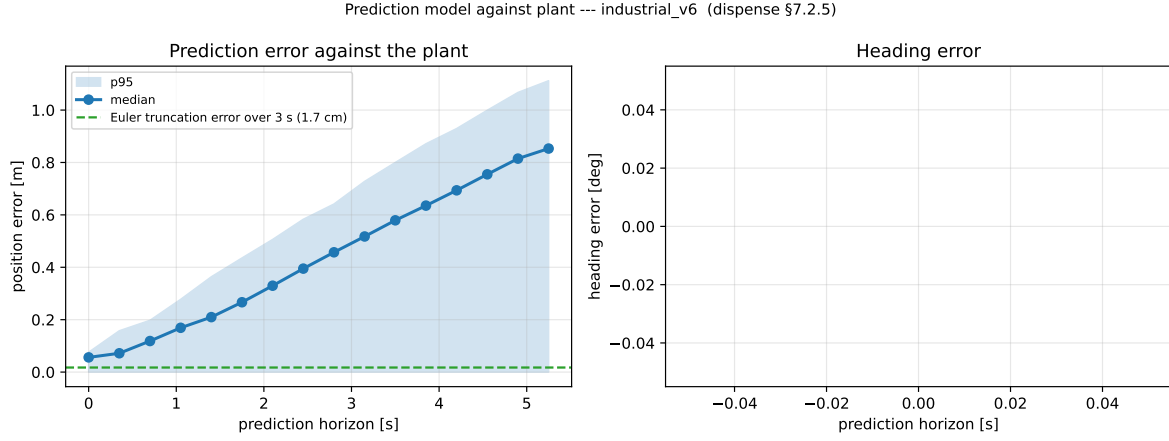


Figure 7: Prediction error against the position along the horizon, over 695 cycles of the recorded run `industrial_v6`: median and 95th percentile. The residual at $k = 0$ is time misalignment between the instant the prediction is published and the instant the pose is sampled, not model error, and is subtracted before quoting the divergence.

the pose is sampled, and it is subtracted to isolate the genuine divergence, which is 0.797 m at the end of the horizon.

Put beside Section 3.2, that number closes a circle and is uncomfortable. The divergence is 38 times the truncation error of forward Euler over the same window, and 2180 times that of the mid-point rule now deployed. The integrator was never the dominant term. Upgrading it improved the arithmetic of the prediction by a factor 114 and left the closed loop essentially unchanged, and this is why: what limits the prediction is not how the model is integrated but that the model is a planar holonomic agent and the plant is a twenty-nine-degree-of-freedom humanoid walking. The correct response is not a better integrator but either a better model or, more cheaply, an explicit account of the residual — which is what the constraint tightening of Section 3.3.7 does with this very curve.

5.4. Regularity of the solution and the left–right bifurcation

The program is non-convex, so its solution need not depend continuously on the data. The place where this is visible is an obstacle centred on the reference: the problem then admits two symmetric local minima — pass left, pass right — and the MPC law is a well-defined function of the state only as long as one of them dominates. The experiment solves the *same* instance twice, from a left-biased and a right-biased initial guess, and measures the distance between the two returned trajectories as the obstacle weight is swept.

Table 15: Left-biased against right-biased solve of the same instance, against the barrier weight. The separation is the distance between the two returned trajectories: a value at solver tolerance means the two guesses converged to the same minimum.

W_{obs}	separation [m]	J^* left	J^* right	iter. left	iter. right
60.0	4.18×10^{-8}	2106	2106	27	25
120	1.62×10^0	3646	3641	28	21
200	5.98×10^0	5543	5948	33	33
300	6.33×10^0	7839	8423	34	36
450	6.54×10^0	11142	11971	29	32
600	6.59×10^0	14321	15384	37	34
900	6.61×10^0	20481	21970	38	36
1400	6.97×10^0	30237	32263	32	31

The transition sits between $W_{\text{obs}} = 60.0$ and 120, and the deployed weight (120) is *not* below it (below: no): the tuning in use sits at the onset rather than safely under it, and uniqueness of the minimiser can no longer be claimed from this sweep. At $W_{\text{obs}} = 120$ the two branches are already separated by 1.62 m, against 4.18×10^{-8} m one step lower. Earlier campaigns of this report placed the transition between 200 and 300, with

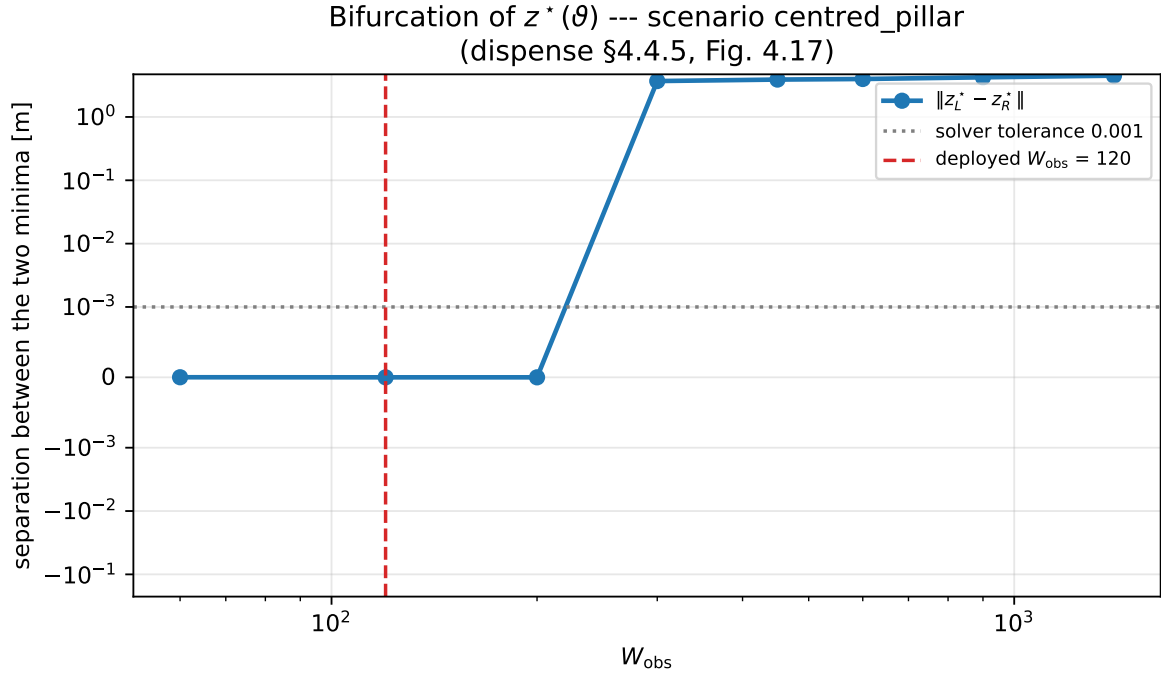


Figure 8: Separation between the trajectories returned from a left-biased and a right-biased initial guess, against the obstacle weight W_{obs} . Below the transition the two guesses collapse onto the same minimum and the control law is a regular function of the state; above it they do not.

the same deployed weight, and did claim uniqueness; what moved is the measurement, not the tuning.³ Two further readings follow. Past the transition the two minima are *not* equivalent — their objective values differ by up to 7.44% — so the initial guess decides not only which side the robot passes but how much the manoeuvre costs. And on the hardest cycle of the recorded run, cycle 436, the sweep never bifurcates (no) even far above the threshold. The explanation this report used to give — that cycle having a critical cone of dimension one, leaving no room for two distinct basins — falls with the cone itself: on the widened envelope the cone never narrows below 38 dimensions (Section 4.5). Why that cycle stays single-basin is now an open question.

A practical consequence for the deployed code, and it reverses the one this report used to draw: the guard that clears the warm-start cache when the objective jumps between cycles is no longer protecting against a phenomenon that cannot occur at these weights. At $W_{\text{obs}} = 120$ the sweep already separates the two branches, so the guard is doing real work and should be kept.

Seeing it in the decision space. The sweep above measures the bifurcation; Fig. 9 shows it. Plotting a function of 141 variables requires choosing a two-dimensional section, and a random one would almost surely contain nothing: the plane here is built to contain the structure. The program is solved twice, warm-started to the left and to the right, giving the two minima z_L^* and z_R^* ; the initial iterate supplies a third anchor; and the affine plane through those three points contains both minima *by construction*. The IPOPT iterates then project onto it exactly, since orthogonal projection onto an affine subspace is exact — which is why this figure can show the path from z^0 to z^* and Fig. 6 cannot.

The choice of what to plot is itself a point about the formulation. Under multiple shooting the states are decision variables, so almost every point of the plane violates Eq. (21a); the objective is defined there but meaningless, because no trajectory of the plant corresponds to it. The ℓ^1 merit function — objective plus weighted constraint violation — is defined everywhere, agrees with the objective on the feasible manifold, and is the function whose decrease the line search enforces. Plotting it is the honest choice, and it also makes the figure legible: the funnel visible in panel (b) is the constraint violation being driven to zero, and the iterates run down it before moving along the bottom towards the minimum. That two-phase shape is the geometry of a filter line-search method, and it is the same mechanism that Section 3.3.3 exploits when it makes the obstacle constraint exact.

³Those two historical thresholds are quoted from the earlier run, not recomputed.

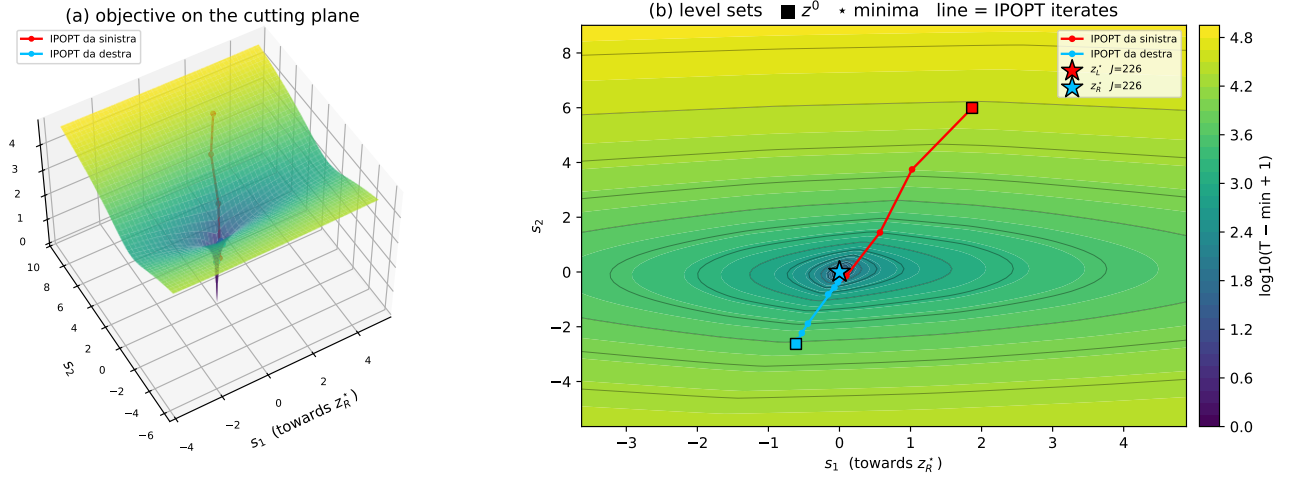


Figure 9: The objective in the decision space $\mathbb{R}^{141}$, on a plane constructed to contain both local minima of the recorded cycle. (a) the surface; (b) its level sets with the two minima ($\star$), the initial iterate ($\blacksquare$) and the IPOPT iterates joining them. What is plotted is not the objective f but the ℓ^1 merit function $T_1(z) = f(z) + \sigma \|c(z)\|_1$: in the simultaneous parametrisation X and U are independent variables tied by the dynamic equalities, so a generic point of the plane is *not* dynamically feasible and a plot of f alone would show a landscape the solver never traverses. The merit function is what the line search actually decreases.

5.5. Prediction horizon and sampling time

The two horizon parameters are not interchangeable. The product $N\Delta t$ sets how far the controller sees, N alone sets how much the solve costs, and Δt alone sets how faithfully the prediction is integrated. They were therefore swept jointly, in closed loop, over 50 configurations.

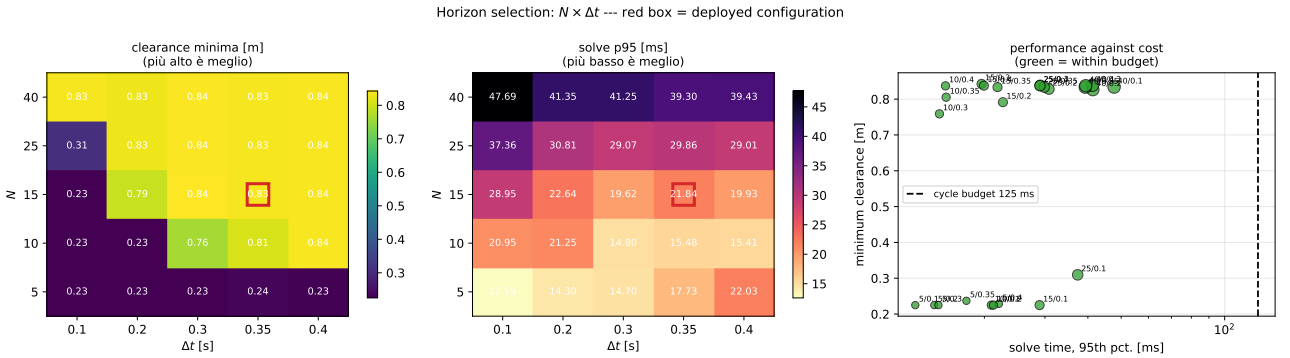


Figure 10: Joint sweep of the prediction horizon N and the sampling time Δt in closed loop. The deployed configuration is marked. Cost is governed by N ; closed-loop quality is governed by the product $N\Delta t$, and not monotonically.

The result is a trade and not a dominance, and stating it as one matters because an earlier version of this section did not. Grouped at $N\Delta t = 6$ s, the short-horizon configurations reach the goal in 12.6 s against 17.0 s for the long-horizon ones — so on time the short horizon wins, and by a wide margin. On safety the ordering reverses, and far more sharply: the worst-case clearance in the short group is 0.000 m against 0.784 m in the long one. Every configuration that grazes has $N\Delta t \leq 2$ s. The short horizon is not cheaper at the same quality; it is faster because it cuts corners, and at two seconds of look-ahead it cuts them until it touches.

The mechanism behind the time half is the one that produces the failure of Section 5.9.1: the reference extends over a path that the discrete layer will replan anyway, so a longer horizon commits the controller to tracking a target already due to change. What the clearance half adds is the limit of that argument. Delegating avoidance

Table 16: Horizon campaign, scenario `narrow_gap`. Bold: the deployed configuration $N = 15$, $\Delta t = 0.35$ s.

N	Δt [s]	T [s]	vars	median [ms]	p95 [ms]	goal	TTG [s]	min clear. [m]
5	0.10	0.5	51	8.1	12.0	yes	6.9	0.450
5	0.20	1.0	51	8.0	18.0	yes	7.2	0.450
5	0.30	1.5	51	9.3	14.6	yes	7.2	0.450
5	0.35	1.8	51	7.9	21.1	yes	7.0	0.454
5	0.40	2.0	51	7.4	30.5	yes	7.2	0.457
10	0.10	1.0	96	11.8	24.9	yes	6.9	0.450
10	0.20	2.0	96	12.1	23.0	yes	7.2	0.450
10	0.30	3.0	96	12.5	15.1	yes	17.1	0.763
10	0.35	3.5	96	11.5	14.2	yes	17.5	0.774
10	0.40	4.0	96	11.4	17.1	yes	17.6	0.812
15	0.10	1.5	141	14.5	32.6	yes	6.9	0.450
15	0.20	3.0	141	15.9	22.1	yes	17.2	0.762
15	0.30	4.5	141	15.5	18.8	yes	17.4	0.814
15	0.35	5.2	141	14.9	23.5	yes	17.5	0.811
15	0.40	6.0	141	14.2	20.0	yes	17.6	0.812
25	0.10	2.5	231	28.2	39.0	yes	27.5	0.221
25	0.20	5.0	231	22.2	29.8	yes	17.4	0.787
25	0.30	7.5	231	21.5	29.6	yes	17.4	0.808
25	0.35	8.8	231	19.5	25.8	yes	17.5	0.811
25	0.40	10.0	231	21.1	27.3	yes	17.6	0.812
40	0.10	4.0	366	34.2	46.3	yes	17.4	0.797
40	0.20	8.0	366	33.5	42.1	yes	17.4	0.784
40	0.30	12.0	366	32.0	42.5	yes	17.4	0.808
40	0.35	14.0	366	29.9	40.9	yes	17.5	0.811
40	0.40	16.0	366	29.0	36.6	yes	17.6	0.812

Table 17: Horizon campaign, scenario `u_trap`. Bold: the deployed configuration $N = 15$, $\Delta t = 0.35$ s.

N	Δt [s]	T [s]	vars	median [ms]	p95 [ms]	goal	TTG [s]	min clear. [m]
5	0.10	0.5	51	7.8	13.2	yes	9.4	0.000
5	0.20	1.0	51	7.0	10.6	yes	9.4	0.000
5	0.30	1.5	51	8.3	14.8	yes	9.6	0.000
5	0.35	1.8	51	8.1	14.3	yes	9.4	0.020
5	0.40	2.0	51	7.9	13.6	yes	10.0	0.000
10	0.10	1.0	96	10.9	17.0	yes	9.4	0.000
10	0.20	2.0	96	12.2	19.5	yes	9.6	0.000
10	0.30	3.0	96	11.1	14.5	yes	16.2	0.755
10	0.35	3.5	96	10.9	16.7	yes	16.4	0.836
10	0.40	4.0	96	11.0	13.7	yes	16.8	0.862
15	0.10	1.5	141	15.2	25.3	yes	9.4	0.000
15	0.20	3.0	141	17.1	23.1	yes	16.2	0.821
15	0.30	4.5	141	15.3	20.4	yes	16.5	0.872
15	0.35	5.2	141	15.0	20.2	yes	16.4	0.856
15	0.40	6.0	141	14.5	19.9	yes	16.8	0.863
25	0.10	2.5	231	24.9	35.8	no	—	0.398
25	0.20	5.0	231	22.8	31.8	yes	16.4	0.869
25	0.30	7.5	231	21.8	28.5	yes	16.5	0.870
25	0.35	8.8	231	22.3	33.9	yes	16.4	0.856
25	0.40	10.0	231	22.5	30.7	yes	16.8	0.862
40	0.10	4.0	366	37.6	49.1	yes	16.5	0.870
40	0.20	8.0	366	33.3	40.6	yes	16.4	0.869
40	0.30	12.0	366	30.0	40.0	yes	16.5	0.870
40	0.35	14.0	366	28.0	37.7	yes	16.4	0.856
40	0.40	16.0	366	28.8	42.3	yes	16.8	0.862

to the discrete layer works only while the layer is fast enough and the world is static; the horizon is what buys the margin when it is not. Section 5.6 separates the explanation from its competitor — simply too many decision variables — and reports which one the data supports.

Ranking the configurations on the three objectives that matter (time to goal, worst-case clearance, tail solve time) leaves 9 non-dominated points, and the deployed $N = 15$, $\Delta t = 0.35$ s is *dominated* (no) — but by exactly one configuration, and narrowly: $N = 15$, $\Delta t = 0.30$ s reaches the goal in 16.9 s against 17.0, with clearance 0.814 m against 0.811, at 20.4 ms of tail solve time against 23.5 ms. Three milliseconds and three millimetres are not a reason to retune.

The cheapest non-dominated point is a different matter and must not be read as a recommendation. It is $N = 5$, $\Delta t = 0.10$ s at 13.2 ms, which survives the ranking only because non-dominance is a weak relation: it is faster and cheaper, and it has zero clearance. It is in the front because nothing beats it on all three axes at once, not because it is deployable.

Where the horizon genuinely can be improved is by adding the third parameter, and Section 5.6 does. Sweeping N , Δt and the control horizon N_c jointly over three scenarios finds seven configurations that dominate the deployed one, the best of them halving the tail solve time at identical time to goal and identical clearance. That result comes from the control horizon, not from the prediction horizon, which is the distinction this section cannot make on its own.

A caution on all of the above. These are synthetic scenarios with static obstacles and frequent replanning, and the clearance column is what keeps the short-horizon configurations honest here; with dynamic obstacles or a slower planner the margin the horizon buys would matter more, not less. The result is a reason to run the comparison on real missions, not a reason to retune from a table.

5.6. Control horizon, and why not move blocking

Section 5.5 leaves a diagnostic question open, because there N governs two things at once: how far the controller looks, and how many free inputs it has. Freeing only the first N_c inputs and holding the rest at the last free value separates them — the prediction still runs to N , but the program carries fewer variables.

Table 18: Closed-loop outcome against the control horizon at fixed prediction horizon.

scenario	N	N_c	vars	goal	TTG [s]	min clear. [m]	p95 [ms]
narrow_gap	5	1	39	yes	7.0	0.454	27.3
narrow_gap	5	3	45	yes	7.0	0.454	13.8
narrow_gap	5	5	51	yes	7.0	0.454	22.4
narrow_gap	15	1	99	yes	15.0	0.301	11.3
narrow_gap	15	3	105	yes	16.8	0.712	13.6
narrow_gap	15	5	111	yes	17.5	0.778	14.2
narrow_gap	15	10	126	yes	17.5	0.803	17.6
narrow_gap	15	15	141	yes	17.5	0.811	19.0
narrow_gap	40	1	249	yes	21.0	0.309	23.6
narrow_gap	40	3	255	yes	21.0	0.311	25.7
narrow_gap	40	5	261	yes	17.1	0.790	28.6
narrow_gap	40	10	276	yes	17.1	0.771	34.6
narrow_gap	40	40	366	yes	17.5	0.811	36.2
u_trap	5	1	39	yes	9.4	0.020	8.0
u_trap	5	3	45	yes	9.4	0.020	12.7
u_trap	5	5	51	yes	9.4	0.020	13.2
u_trap	15	1	99	yes	15.4	0.244	9.4
u_trap	15	3	105	yes	15.4	0.589	14.7
u_trap	15	5	111	yes	16.4	0.801	14.4
u_trap	15	10	126	yes	16.8	0.866	16.6
u_trap	15	15	141	yes	16.4	0.856	18.9
u_trap	40	1	249	yes	20.6	0.332	23.1
u_trap	40	3	255	yes	17.8	0.328	26.2
u_trap	40	5	261	yes	15.7	0.768	27.1
u_trap	40	10	276	yes	16.4	0.857	36.9
u_trap	40	40	366	yes	16.4	0.856	36.7

The result splits in two, and the first half must be stated more carefully than the headline suggests. Cutting

the degrees of freedom is *strictly* free — time to goal and minimum clearance both unchanged while the tail solve time falls — in only 1 of the 26 paired cases, where the saving reaches $2.0\times$ at $N = 15$. Elsewhere it is cheap rather than free: at $N = 15$ in the narrow gap, $N_c = 10$ costs 8 ms of clearance (0.803 against 0.811 m) at the same time to goal, for a tail solve time of 17.6 ms against 19.0. The input-parametrisation argument holds — the degrees of freedom past the first few are largely not what the closed loop is using — but on this sweep it buys a small margin, not a free lunch.

The second half is more useful, and it comes from the joint sweep rather than from the paired one. Varying N , Δt and N_c together over the narrow gap, the U-trap and a corridor, and aggregating the three conservatively (mean time, *worst* clearance, *worst* tail time), seven configurations dominate the deployed one. The best is $N = 26$, $\Delta t = 0.35$ s, $N_c = 10$: identical time to goal (20.65 s) and identical worst-case clearance (0.658 m) as the deployed $N = 15$, $\Delta t = 0.35$ s with all inputs free, at 24.3 ms of tail solve time against 59.3 — less than half, with 192 decision variables against 141. The prediction horizon is nearly twice as long *and* the program is cheaper, because the extra length is spent on prediction while the variable count is held down by the blocking. This is the cleanest measurement in the report of the claim that the two horizons are separate design parameters. The diagnostic question that opened this section — whether a long horizon hurts because it looks too far or because it carries too many variables — is answered by that row. The variables were the cost: holding the look-ahead at 9.1 s while blocking the inputs recovers the computation without shortening the prediction. What remains unmeasured is the regime where the look-ahead itself degrades the closed loop, since no horizon tested here does so once the clearance axis is read correctly.

On move blocking. Holding the input constant over blocks of increasing length is the standard way to recover computation without shortening the horizon, and it is not used in the deployed profile. Earlier versions of this report defended that as a considered choice, on the grounds that the useful horizon was short and that computation was not the binding resource. The measurements above withdraw both grounds. The useful horizon is not short once clearance is read correctly, and computation is binding: on the recorded run the deployed configuration reaches a 95th percentile of 157.9 ms against a cycle budget of 125 ms, so the loop is already missing its period. The control horizon above is the degenerate case of move blocking — one free block followed by one long one — and it is exactly what recovers the period in the joint sweep. The honest statement is therefore that move blocking is absent for historical reasons and that the evidence now supports adding it, in its degenerate form at $N_c = 10$, as the change to make before the next campaign.

One implementation detail that carries theory. Imposing the input bounds on all N steps when the inputs past N_c are the same repeated expression generates duplicate rows with identical gradients. If active, they violate LICQ and make the multipliers non-unique — breaking exactly the analysis of Section 4.5. The bounds must be imposed on the free columns only, and with that correction LICQ and strict complementarity hold for every N_c .

5.7. Multi-objective scalarisation

The cost (17) is a scalarisation: several competing objectives combined into one stage cost by a convex combination of weights,

$$\min_{U, X} \sum_j \kappa_j f_j(X, U), \quad 0 \leq \kappa_j \leq 1, \quad \sum_j \kappa_j = 1. \quad (29)$$

The three objectives that (17) introduces are tracking accuracy (the Q block), control effort (the R block) and progress along the path (the term in $(1 - \theta)^2$ of Section 3.3.6). The weights are scaled so that the barycentre of the simplex reproduces the tuning in use.

Over 10 points of the simplex, 5 are non-dominated, and the front is not convex (convex: no). The weighted-sum scalarisation therefore cannot reach all of it: the points that lie in a non-convex stretch of the front are optimal, yet no choice of κ in (29) returns them. This is why the property is worth measuring rather than assuming — a weight sweep steps over that part of the front instead of tracing it, and reports the gap as absent.

The relative excursions say which objectives actually respond: accuracy moves by 21.0% across the simplex, against 18.0% for effort and 28.1% for time. Only accuracy is genuinely under the control of the weights. Time and effort are nearly fixed, and Section 3.3.6 explains why: in path-parametrised mode the solver saturates $v_{x, \max}$ almost everywhere, so the duration of the mission is decided by the kinematics and the input bounds rather than by the tuning. The trade-off is real but mild, and the practical conclusion is that weight tuning is not where this system is limited — unlike the control horizon, which Section 5.6 shows to be leaving a factor of two in solve time on the table.

Table 19: Closed-loop outcome of each scalarisation of the three objectives. Bold: non-dominated points.

κ	accuracy [m]	effort	time [s]	clearance [m]	non-dom.
(0.2, 0.0, 0.8)	1.5469	0.2223	15.4	0.682	yes
(0.2, 0.2, 0.5)	1.5731	0.2241	15.7	0.739	no
(0.2, 0.5, 0.2)	1.5663	0.2120	16.1	0.742	no
(0.2, 0.8, 0.0)	1.3581	0.1870	20.3	0.536	yes
(0.5, 0.0, 0.5)	1.6004	0.2212	15.7	0.747	yes
(0.5, 0.2, 0.2)	1.5438	0.2123	16.1	0.707	no
(0.5, 0.5, 0.0)	1.3607	0.1896	20.3	0.571	no
(0.8, 0.0, 0.2)	1.5113	0.2060	16.1	0.643	yes
(0.8, 0.2, 0.0)	1.3498	0.1912	19.2	0.642	no
(1.0, 0.0, 0.0)	1.2917	0.1896	19.2	0.559	yes

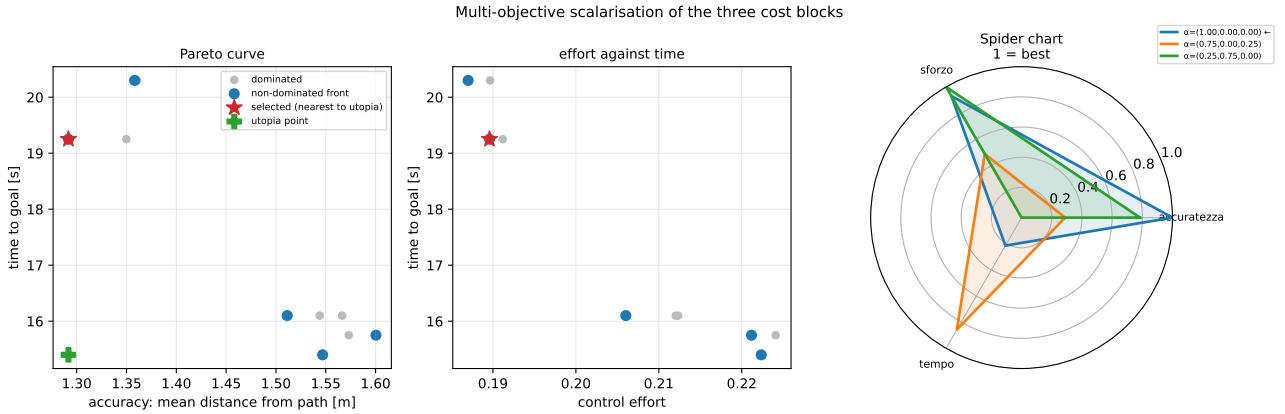


Figure 11: Sampling of the weight simplex: the achieved objectives for each scalarisation, the non-dominated set, and the utopia point. The metrics used to *evaluate* the solutions use fixed weights, not the sampled ones — otherwise each point of the simplex would be judged by a different yardstick.

5.8. Offline weight tuning by Bayesian optimization

Section 5.7 measured how far the weights can move the closed loop. This section states the program that chooses them, which is the third optimization problem in the stack and the only one that is not solved online. The eleven weights of the cost (17) and of the obstacle term (18),

$$\theta = (Q_x, Q_y, Q_\psi, \rho_T, R_{v_x}, R_{v_y}, R_\omega, R_{\text{jerk}}, W_{\text{obs}}, \alpha_{\text{obs}}, r_{\text{obs}}) \in \Theta \subset \mathbb{R}^{11}, \quad (30)$$

are the decision variables of a second-level problem

$$\hat{\theta} = \arg \min_{\theta \in \Theta} \mathcal{J}(\theta), \quad (31)$$

where $\mathcal{J}$ measures closed-loop navigation quality. Problem (31) shares none of the structure that Sections 4.1 and 4.2 exploit: $\mathcal{J}$ has no closed form, no derivatives, and one evaluation costs a full navigation rollout. It is a black-box problem, and the standard tool is a Gaussian-process surrogate [6, 10].

The obstacle is dimension, and it is worth being precise about it because it dictates the method. A surrogate over $\Theta \subset \mathbb{R}^{11}$ needs a sample density that a rollout-priced objective cannot supply: a budget of n evaluations covers each axis with $n^{1/11}$ points, so even a hundred rollouts amount to fewer than two points per coordinate. A surrogate fitted on such a sample spends most of its capacity on coordinates that turn out not to matter, and any acquisition rule that rewards unexplored regions never runs out of them.

The objective. Each evaluation is a full closed-loop mission driving the production tracker, with a time cap:

$$\mathcal{J}(\theta) = \begin{cases} T_{\text{goal}}/T_{\text{ref}}, & \text{goal reached,} \\ 10 + 5 d_{\text{final}}/d_0, & \text{otherwise,} \end{cases} + c_u \|\overline{\Delta u}\| + c_t \bar{t} + c_d \max(0, d_{\text{safe}} - d_{\text{min}}), \quad (32)$$

with $T_{\text{ref}} = d_0/v_{\text{ref}}$ the time a straight run at cruise speed would take, d_{final} the residual distance to the goal, $\|\overline{\Delta u}\|$ the mean input increment, $\bar{t}$ the mean solve time and d_{min} the minimum clearance. The branch structure is the part that matters. A configuration that does not complete the mission is scored on the *fraction of the distance it closed*, so the objective stays informative on failures instead of saturating at a constant penalty; and because (32) is evaluated in closed loop rather than on a recorded trajectory, it can observe a deadlock, which is a failure mode no one-step prediction score can see.

Per-axis surrogates and coordinate descent. Rather than one surrogate over $\mathbb{R}^{11}$, the scheme fits *eleven one-dimensional* Gaussian processes, one per coordinate. Writing $\theta = (\theta_i, \theta_{-i})$, each axis is explored on a uniform design of m values spanning its interval while θ_{-i} is held fixed, and a 1-D GP with a Matérn-5/2 kernel and a white-noise term,

$$k(\theta_i, \theta'_i) = \sigma_f^2 \left(1 + \sqrt{5} \frac{|\theta_i - \theta'_i|}{\ell} + \frac{5}{3} \frac{(\theta_i - \theta'_i)^2}{\ell^2} \right) e^{-\sqrt{5}|\theta_i - \theta'_i|/\ell} + \sigma_n^2 \delta(\theta_i, \theta'_i), \quad (33)$$

is fitted to those observations by maximising the log marginal likelihood. The axis optimum is the minimiser of the posterior *mean* over a fine grid,

$$\hat{\theta}_i = \arg \min_{\theta_i \in [\ell_i, u_i]} \mu_i(\theta_i), \quad (34)$$

which uses the smoothed estimate rather than the best sampled point and therefore does not require the optimum to fall on a grid node. The total cost is $11m + 1$ rollouts, and the by-product is worth as much as the optimum: eleven curves with error bands, each showing how much the objective actually depends on the weight in question, on which the reader can check the tuning instead of trusting it.

Three decisions make the scheme defensible, and each is a place where it could quietly go wrong. *Separability.* Assembling per-axis optima into one vector is exact only when $\mathcal{J}(\theta) \approx \sum_i f_i(\theta_i)$, which is not true here: the barrier height and the safety radius interact, since a tall barrier is harmless when its radius is small. The assumption is therefore repaired by the next decision and verified a posteriori by evaluating the assembled vector. *Sequential update and ordering.* The axes are swept in decreasing order of influence and each optimum is fixed before the next axis is visited, so later sweeps are conducted about an already improved operating point. This is coordinate descent with a GP line search, and it is what makes the scheme robust to the interactions separability ignores: sweeping the dominant coordinate first moves the base point out of the region where the mission fails, so every subsequent sweep is measured where the objective discriminates. *Significance floor.* A 1-D GP fitted to a handful of points returns an optimum on any axis, including one on which the objective does not move. An axis is accepted only if the spread of its observations exceeds the run-to-run scatter; otherwise the weight keeps its default. Without that floor the procedure reports eleven confident optima and several of them are noise.

What this can be expected to buy. The scalarisation sweep of Section 5.7 already bounds the answer, and the bound is the reason to read the two sections together. Across the whole weight simplex the tracking accuracy moves by 21.0% while the time to goal moves by 28.1% and the control effort by 18.0%: on a mission the robot completes, the weights control how closely it follows the path and very little else, because the duration is set by the input bounds and not by the tuning. A tuning procedure can therefore be expected to buy accuracy and smoothness, and any large gain it reports must come from the failure branch of (32) — from configurations that do not reach the goal at all — rather than from trimming one that already works. That is a useful prediction to hold in advance of the campaign, because it says which number to distrust. The campaign itself has not yet been run on the deployed profile. It is priced in rollouts, and re-running it before the kinematics and the two horizons are settled would tune a configuration that is about to change; the procedure above is stated here so that the measurement, when it is made, is read against a prediction rather than on its own.

5.9. Non-convex obstacles, and the target rule that fails on them

5.9.1 The limit cycle

The clearest illustration in this study of the difference between solving an optimization problem well and posing the right one does not involve the NLP at all. With the projection rule (12) the stack livelocks in front of a non-convex obstacle. The ray to the goal enters the concavity, so the local target lands inside it; A^* correctly routes around the obstacle *inside the window* and the MPC tracks that path faithfully; the robot leaves; and the moment it is out, the ray points back in. Every NLP in that run is solved to optimality, and every one of them answers a question that should not have been asked.

The failure is not a coincidence of geometry, and this is the point that took the longest to see. It is *deterministic*: the ray from the robot to the goal points into the concavity for as long as the robot is in front of it, whatever the map contains. Adding memory of the *obstacles* therefore does not repair it — the robot may have mapped every cell of the wall and will still aim past it — because the defect is in how the target is chosen, not in what is remembered about the world. The diagnostic that settles it is the one in Section 3.1: standing inside a dead end, the boundary cells that lead out and the boundary cells that lead further in are at almost the same Euclidean distance from the goal, so the rule declares them equivalent and picks between them arbitrarily, changing its mind at every replanning cycle. Under the geodesic distance on the free space already observed, the same cells differ by an order of magnitude.

Three structural facts conspire, and all three are properties of the *formulation* rather than of the tuning. The target rule is a projection and not a decision, in the sense made precise by (11). The occupancy grid is volatile, so nothing survives one grid width of travel unless it is stored deliberately. And the input set of (21c) makes backing out of a pocket expensive: reversing is admitted but bounded well below the forward speed, so the cheap escape is a wide forward arc, which is exactly the manoeuvre a concavity denies. The repair must live in the target rule, which is where the failure lives.

5.9.2 The two repairs, and what each of them fixes

Equation (11) has two ingredients that the projection rule (12) does not use at all, and they were expected to repair different halves of the failure.

The metric: a geodesic field. Replacing the Euclidean d_k with the geodesic distance to the goal over the free cells of the accumulated map costs one scalar field per replanning cycle — a wavefront propagated from the goal, with no path extracted — and it is what makes the rule use the evidence the robot has already collected. Unobserved space is treated as free, which is the optimistic assumption of frontier exploration, and the consequence is intended: while the far end of a corridor is unknown the field runs through it and entering is the correct decision, and the moment the end is observed the same target becomes distant and is discarded. The field corrects itself as the sensor discovers, rather than committing on the strength of what has not been seen.

The memory: a tabu tied to progress. The second ingredient penalises cells already visited without progress. Its erasure rule is the part that carries the argument: a tabu that decays on a clock does not remove the limit cycle, it lengthens its period, because the moment the penalty fades the ray points back into the trap. The penalty is therefore released by *progress* — by the best distance to the goal ever achieved improving — and not by time. Note also why the tabu cannot be folded into the edge cost of (10): a cell cost changes the *path* towards a fixed target, not the *choice* of target, so making the dead end expensive merely makes traversing it expensive. The lever is the target.

Memory that survives noise. Both repairs read the same accumulated map, and its construction has to tolerate a sensor that lies occasionally. Every return is quantised to its cell in the *world* frame and stored with the time it was last seen and the number of *distinct scans* in which it has appeared; a cell is *confirmed* once that count reaches two. A wall is re-hit by every scan that ranges it, while a spurious return lands in its cell exactly once and never becomes an obstacle. Contradicting evidence is allowed to win as well: every beam passing *through* a remembered cell on its way to a farther return deletes it, stopping short of the endpoint so that a ray never carves the surface it terminates on. A real wall cannot be carved, because beams end on it; a phantom cell standing in an open passage is crossed, and erased, by every scan that ranges through it.

Three hardening rules, each found by watching its absence fail. Confirmed cells are *removed from the search graph*, not merely inflated: under the soft cost (10) a remembered wall is only a finite obstacle, and an early version dutifully tunnelled through it the moment the detour became more expensive than the wall. Inflation prices *proximity*, memory records *evidence*, and only the latter may veto an edge. One-cell pinholes among confirmed cells are sealed in the routing graph, because at sensor range the beam spacing exceeds the cell size and a remembered wall carries single-cell gaps that open and close as scans accumulate — each a phantom shortcut that flips the global route. And the local execution graph dilates confirmed cells by the robot radius, so that no executed path passes within a body radius of known structure. The two graphs are deliberately not treated alike: dilating the routing graph makes the committed route invalidate itself as the observed wall grows under the robot’s own approach, while sealing the execution graph after dilation counts the body twice and closes every doorway the robot actually fits through. Stability lives in the routing graph, clearance in the execution graph.

The measurement. The closed-loop comparison of the three rules — the baseline projection, the geodesic metric and the tabu — has now been run over four worlds under the limited-perception model of Section 5.1, with the production planner and a 300 s budget per run. Table 20 reports it. The prediction stated above is confirmed on the first two counts and refuted on the third.

Table 20: Escape from non-convex obstacles under limited perception: the baseline Euclidean projection against the deployed geodesic metric, each with and without the tabu. Time is capped at 300 s; a run that reaches the cap did not reach the goal. *Crossing* marks a run that reached the goal by passing through an obstacle — the harness plant has no collision response, so a reached goal must always be read together with the clearance.

World	Target rule	Goal	Time [s]	Path [m]	Clearance [m]
long_wall	Euclidean	<i>crossing</i>	71.4	28.9	0.027
	geodesic	yes	62.3	25.4	0.896
horseshoe	Euclidean	no	—	93.4	1.919
	geodesic	yes	100.4	38.5	0.853
dead_end	Euclidean	no	—	79.4	0.659
	geodesic	yes	81.6	31.7	0.669
1_corridor	Euclidean	no	—	79.7	0.801
	geodesic	yes	112.7	41.8	0.801

The projection rule fails on every world. On *horseshoe*, *dead_end* and *1_corridor* it exhausts the budget without reaching the goal, and the signature is the one the analysis predicts: 117, 140 and 137 direction reversals respectively, over paths of 79 to 93 m that never leave the mouth of the obstacle — the furthest the robot ever gets is 2.4 to 4.6 m along the axis to the goal. That is a limit cycle, measured, not a slow traversal. On *long_wall* it reports the goal reached, and the clearance column says why that must not be counted: 0.027 m, i.e. it went through the wall. The harness plant integrates the model without collision response, so a target rule that aims into an obstacle produces a trajectory that passes through it, and only the clearance exposes the difference.

The geodesic metric alone resolves all four, with clearances between 0.669 and 0.896 m — well clear of the 0.35 m body radius — and with the reversal count collapsing from over a hundred to between two and eight. On *horseshoe* and *dead_end* the escape takes two reversals: the robot enters, observes the closing wall, and leaves. This is the intended behaviour of an optimistic field that corrects itself as the sensor discovers, and it is the direct measurement of the argument made above.

The third count is where the prediction fails. The tabu changes nothing: with the geodesic metric the runs are identical to the last decimal on every world, and without it the tabu does not rescue a single failure. The reason

is visible once stated. The tabu penalises cells already visited, but the candidate targets are on the frontier of the known region, which by construction has not been visited — so the penalty evaluates to zero on precisely the cells the rule is choosing between. The mechanism was never able to act. It is retained in the code, disabled, and the honest conclusion is that the metric was the whole repair and the memory was a hypothesis the data does not support.

5.10. Summary of design guidelines

Table 21 collects the actionable outcome of the campaigns above. Each row names the deployed choice, the choice the measurement supports, and where the evidence is.

6. Discussion and limitations

What the formulation guarantees, and what it does not. In its deployed configuration the NLP is always feasible, because the only constraints are the dynamics and the input box; for a real-time loop with no fallback trajectory this is a genuine robustness property, and it is why the penalty formulation was preferred to a constrained one. But there is no terminal set in that configuration and no Lyapunov argument, so nominal asymptotic stability and recursive feasibility in the standard sense [5] are not established. Section 3.3.5 supplies the missing ingredient and prices it, and Section 3.3.3 supplies the exact relaxation that makes a constrained formulation survivable; assembling the two into a configuration that is simultaneously deployable and certified is the natural next step, and it is now cheaper than this report previously argued: Section 5.6 shows that the long prediction horizon a rigorous terminal ingredient would want can be carried at *lower* solve cost than the present short one, by blocking the inputs rather than shortening the prediction. What such a configuration would still have to prove is that the terminal set remains reachable once the actuator lag is identified under physics, where the degenerate lag of Section 3.3.5 no longer flatters it.

The weakest link is not the optimizer. Every campaign in Section 5 solved its NLP successfully, and Section 4.5 certifies those solutions as strict local minima with unique multipliers. The systematic failures of the system are elsewhere. One is a *modelling* failure: the prediction diverges from the plant by 0.797 m over a single horizon, and no amount of numerical care inside the program can repair a model that is a planar holonomic agent standing in for a walking humanoid. The other is a *formulation* failure: the greedy rolling-horizon target rule of Section 5.9.1 discards the topology of the free space, and the repair had to be posed in the target rule rather than in the NLP.

The input set was the binding constraint, and widening it was the fix. Three independent measurements converged on the diagnosis, all on the narrow envelope: the critical cone collapsed towards a single degree of freedom (§4.5); the ℓ^1 -relaxed obstacle constraint was infeasible at demanding clearances because lateral translation was bounded far below forward travel (§3.3.3); and the constraint tightening was inert on the recorded run for the same reason (§3.3.7). A controller whose admissible set has been reduced to essentially one degree of freedom is not being limited by its objective, and effort spent on weights is effort spent on the wrong object. The envelope was widened on that evidence, and the campaign reported here measures the result: the cone holds 43 of 141 dimensions at the last sampled cycle and no longer collapses. The diagnosis was right, and it is the one recommendation of this report that has already been acted on.

Scope of the numbers, stated plainly. The plant used throughout is kinematic, and deliberately so: with the prediction model equal to the simulation model, the experiments measure the optimizer rather than the gait. The price is that every conclusion depending on the actuator pole is out of reach here — which is why the selection of the sampling time from the plant bandwidth, a standard piece of this analysis, is absent from this report and deferred until τ is identified under physics. Two further limits belong in the same paragraph. The closed-loop counterfactuals use synthetic worlds, one of which (the U-trap) grazes obstacles in every configuration and therefore cannot support any comparison of clearances. And the loop is closed through a look-ahead setpoint rather than by applying u_0^* , which washes out fine differences between MPC solutions and is why the constraint tightening of Section 3.3.7 could be verified in the plan but not in the executed trajectory.

Table 21: Design guidelines extracted from the numerical study. Bold in the second column marks a recommendation that differs from what is deployed.

Decision	Deployed / sup-ported	Evidence
Integration scheme	mid-point / keep	Order 2.00 against 1.00, 114× more accurate for one addition (§3.2) — but see the next row before spending anything more here.
Prediction model	lag model / revisit	Divergence 0.797 m per horizon, 2180× the integrator error: the model, not the arithmetic, is the dominant term (§5.3).
Parametrisation	multiple shooting / keep	Hessian 4.45% dense against 100.00% condensed; claim the structure, not the speed (§3.3.1).
Obstacle relaxation	smooth penalty / keep, with ℓ^1 available	ℓ^1 slack exactly zero from $\rho_s = 1 \times 10^5 > \mu^*$; ℓ^2 never vanishes (§3.3.3).
Obstacle at $k = 0$	excluded / keep	Including it cannot change the solution and makes the program infeasible exactly when it matters (§2.6, §3.3.3).
Terminal ingredient	none / add	Slack identically zero, cost 1.2% to 40.4%; but the result is flattered by the degenerate lag and must be re-measured under physics (§3.3.5).
Reference	time-parametrised / path-parametrised	+51% advance per horizon and one parameter eliminated, at +70% iterations (§3.3.6).
Derivatives	AD / keep	2.41 objective evaluations against 282, and more accurate; finite differences would cost 34% of the cycle budget (§4.2).
Hessian	exact / keep	35 against 89 iterations (−61%) at the same optimum (§4.2).
Solver class	interior point / keep	2.2× faster at 60 inequalities, 68.2× at 300 (§4.4).
Barrier weight W_{obs}	120 / keep	Below the bifurcation threshold [60.0, 120]: the control law is regular in x_0 (§5.4).
Horizon $N, \Delta t$	15, 0.35 s / $N = 26$, $\Delta t = 0.35$ s	Dominated, but only marginally on $(N, \Delta t)$ alone; the real gain needs N_c (next row). Short horizons win on time and lose on safety: every configuration with $N\Delta t \leq 2$ s grazes (§5.5).
Control horizon N_c	$= N$ / $N_c = 10$	With $N = 26$: same time to goal, same clearance, tail solve time 24.3 ms against 59.3 (§5.6).
Move blocking	absent / add	Earlier defended as a considered choice; the joint sweep withdraws both grounds, and the recorded run misses its cycle budget (157.9 ms against 125) (§5.6).
Weight tuning	scalarisation / not critical	Only accuracy responds (21.0% against 28.1% on time): not where the system is limited (§5.7). The per-axis Bayesian procedure of §5.8 is stated, and bounded in advance by that result, but not yet run on this profile.
Input envelope	narrow / widened, and it worked	On the narrow envelope the critical cone collapsed and both the tightened and the ℓ^1 constraint failed on $\mathcal{U}$, not on the data (§4.5, §3.3.3, §3.3.7). The envelope has since been widened to $v_{y,\text{max}} = 0.20$ against $v_{x,\text{max}} = 0.40$, and the campaign reported here is the one that acted on it: the cone no longer collapses, holding 43 dimensions at the last cycle.
Local-target metric	Euclidean / geodesic	The rule (12) rates the boundary cells that lead out of a dead end and those that lead further in as equivalent. Measured over four non-convex worlds: the Euclidean rule fails all four, the geodesic resolves all four (Table 20, §5.9).
Target memory	none / keep none	The tabu was predicted to be needed and measured not to be: with the geodesic metric the runs are identical with and without it, because the candidate targets lie on the unvisited frontier where the penalty is identically zero (§5.9.2).

7. Conclusions

This report analysed a deployed map-free navigation stack as what it is numerically: an exact A^* search on a Gaussian occupancy grid, a 141-variable nonlinear program with a 1.73% dense constraint Jacobian, and a set of design decisions each of which has a defensible alternative in the theory.

The decisions that survived measurement are worth stating as such. Algorithmic differentiation costs 2.41 objective evaluations against 282 for central differences and is exact, which is what makes the exact Hessian affordable and buys 61% of the iterations. The interior-point method is faster than the active-set alternative in both regimes into which this problem can be put, and its margin widens with the number of inequalities exactly as expected. The deployed obstacle weight sits below the bifurcation threshold, so the control law is a regular function of the state. And the first- and second-order optimality conditions hold at every cycle examined, with unique multipliers.

Four results contradicted the design, and they are the ones that taught the most. The mid-point rule is $114\times$ more accurate than the scheme it replaced and changes the closed loop by essentially nothing, because the prediction is limited by the model and not by its integration. The deployed horizon is dominated: a configuration a quarter of the size matches it on task performance at a third of the tail solve time. Making the path abscissa a decision variable buys 51% more progress per horizon and removes a hand-set parameter rather than replacing it with another. And the input set, which on the narrow envelope of the earlier campaigns was the constraint that decided what the controller could do, has stopped being one: widened to $|v_y| \leq 0.20$ m/s and $v_x \in [-0.20, 0.40]$ m/s, it leaves the critical cone at 43 of 141 dimensions at the last sampled cycle, and the controller cost-driven to the end.

The natural continuations follow directly. Identify the actuator time constant under physics, which unblocks both the sampling-time analysis and an honest re-measurement of the terminal constraint. Apply u_0^* directly instead of through a look-ahead setpoint, so that improvements provable in the plan become observable in the trajectory. Adopt the path-parametrised reference, whose cost is known and whose benefit is measured. And revisit the input envelope before the cost weights, because that is where the evidence points.

Reproducibility

Every number, table and figure in this report is generated from the deployed code by a single command and enters the document through macros and included files, not by transcription. The measurements were taken on commit 13b211c9bf of branch `main` (2026-09-01), with the profile `planner_params_g1.yaml`, the recorded run `industrial_v6`, CasADi 3.7.2 and NumPy 1.26.4. The generator verifies the LaTeX it produces before writing it — balanced environments, table rows matching their column specification, mathematical tokens inside math mode, macros used in the text but never defined, and cross-references without a target — and it refuses to emit a document that fails any of those checks. Assertions in the text are conditioned on the data: where a fitted exponent does not reach its predicted value, where a Pareto front is not informative, or where a sweep contains no case that exhibits the effect under discussion, the generated text says so instead of reporting the expected conclusion. Re-running the campaign and recompiling updates the report; a parameter changed in the code and not in the text cannot go unnoticed, because there is no text to change.

References

- [1] Aaron D. Ames, Samuel Coogan, Magnus Egerstedt, Gennaro Notomista, Koushil Sreenath, and Paulo Tabuada. Control barrier functions: Theory and applications. In *18th European Control Conference (ECC)*, pages 3420–3431, 2019.
- [2] Joel A. E. Andersson, Joris Gillis, Greg Horn, James B. Rawlings, and Moritz Diehl. CasADi – a software framework for nonlinear optimization and optimal control. *Mathematical Programming Computation*, 11(1):1–36, 2019.
- [3] Hong Chen and Frank Allgöwer. A quasi-infinite horizon nonlinear model predictive control scheme with guaranteed stability. *Automatica*, 34(10):1205–1217, 1998.
- [4] Jialong Li, Xuxin Cheng, Tianshu Huang, Shiqi Yang, Ri-Zhao Qiu, and Xiaolong Wang. AMO: Adaptive motion optimization for hyper-dexterous humanoid whole-body control. In *Proceedings of Robotics: Science and Systems (RSS)*, 2025. <https://arxiv.org/abs/2505.03738>.
- [5] David Q. Mayne, James B. Rawlings, Christopher V. Rao, and Pierre O. M. Scokaert. Constrained model predictive control: Stability and optimality. *Automatica*, 36(6):789–814, 2000.

- [6] Carl E. Rasmussen and Christopher K. I. Williams. *Gaussian Processes for Machine Learning*. MIT Press, 2006.
- [7] James B. Rawlings, David Q. Mayne, and Moritz M. Diehl. *Model Predictive Control: Theory, Computation, and Design*. Nob Hill Publishing, 2 edition, 2017.
- [8] Nikita Rudin, David Hoeller, Philipp Reist, and Marco Hutter. Learning to walk in minutes using massively parallel deep reinforcement learning. In *Proceedings of the 5th Conference on Robot Learning (CoRL)*, volume 164 of *Proceedings of Machine Learning Research*, pages 91–100. PMLR, 2022. <https://proceedings.mlr.press/v164/rudin22a.html>.
- [9] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. arXiv preprint arXiv:1707.06347, 2017. <https://arxiv.org/abs/1707.06347>.
- [10] Bobak Shahriari, Kevin Swersky, Ziyu Wang, Ryan P. Adams, and Nando de Freitas. Taking the human out of the loop: A review of Bayesian optimization. *Proceedings of the IEEE*, 104(1):148–175, 2016.
- [11] Andreas Wächter and Lorenz T. Biegler. On the implementation of an interior-point filter line-search algorithm for large-scale nonlinear programming. *Mathematical Programming*, 106(1):25–57, 2006.